

Mountain Village Cooperative Fall Season Drift Gillnet Test Fishery Project, 2022

by

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Final report to the U.S. Research and Management Fund
Report Number: RM-10_22

June 2023

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ABSTRACT

Test fishing activities at the Mountain Village Test Fish (MVTF) sites were conducted from July 18 through September 12, 2022, inclusive. Of the total possible season test fishing drifts, 228, 73.7% were successfully conducted. Test fishing was conducted at Site 5 for the entire season. However, test fishing was delayed at Site 6 because of high water conditions and was initiated during the evening drift period on August 10. The delayed test fishing at Site 6 resulted in most of the successful test fish drifts, 61.3%, occurring at Site 5. Although test fishing at Site 6 accounted for only 38.7% of the successful test fishing drifts, a majority of the Fall Chum, *Oncorhynchus keta*, 67.8% and Coho Salmon, *O. kisutch*, 68.3% were caught at Site 6.

The total Fall Chum Salmon MVTF catch in 2022 for Catch per Unit Effort (CPUE) analysis totaled 152 salmon. Included in this total were 10 Fall Chum Salmon that were observed to fall out of the net while it was being retrieved. The total cumulative MVTF CPUE for Fall Chum Salmon totaled 607.00. Additionally, three Fall Chum Salmon were caught but not included in the CPUE analysis but were sampled for ASL analysis. Of the total number of Fall Chum Salmon caught, 155 salmon, 145, were sampled for Age, Sex, and Length (ASL). Of the 145 salmon sampled, 137 Fall Chum Salmon, or 94.5%, were successfully aged. All sampled Fall Chum Salmon sampled were successfully sexed and measured. Female Fall Chum Salmon comprised 57.7% of the aged sample, 75.0% of the unaged sample, and 58.6% the total sample. The mean length of all measured Fall Chum Salmon was 563 mm and aged Fall Chum Salmon was also 563 mm. The total and the aged sample of male Fall Chum Salmon averaged 564 mm. The total and the aged sample of female Fall Chum Salmon averaged 562 mm in size. Most, 74.5%, of the sampled Fall Chum Salmon were age-0.3 fish, followed by age-0.4 salmon, 24.1% and age-0.2, 1.5%. Mean length by age class was 510 mm (n=2) for age-0.2, 561 mm (n=102) for age-0.3, 571 mm (n=33) for age-0.4 fall chum salmon. No age 0.5 fish were sampled. The midpoint of the run occurred on August 17. The mid-50% of the run occurred from August 9 to August 23. Based on the historic data, the Fall Chum Salmon run was approximately 8 days later and 33.3% compressed than the historic median run.

The total Coho Salmon MVTF catch in 2022 for CPUE analysis totaled 82 fish. Included in this total was three Coho Salmon that were observed to fall out of the net while it was being retrieved. The cumulative MVTF CPUE for Coho Salmon totaled 259.0. Additionally, one Coho Salmon was caught but not included in the CPUE analysis but was sampled for ASL analysis. Of the 83 Coho Salmon caught, 80 Coho Salmon were sampled for ASL. Of the total number sampled, 72 Coho Salmon, or 90.0%, were successfully aged. All sampled Coho salmon sampled were successfully sexed and measured. Female Coho Salmon comprised 38.9% of the aged sample, 62.5% of the unaged sample, and 41.3% of the total sample. Mean length of Coho Salmon was 538 mm for the aged sample, 534 mm for the unaged sample, and 538 mm for the total sample. Male Coho Salmon averaged 540 mm for the aged sample, 522 mm for the unaged sample, and 539 mm for the total sample. Female Coho Salmon averaged 536 mm for the aged sample, 541 mm for the unaged sample and 537 mm for the total sample. Age-2.1 dominated the sample, accounting for 76.4% of the aged sample. Age -1.1 Coho Salmon comprised 20.8% of the sample. Age-3.1 coho salmon accounted for 2.8% of the sample. Mean length of age-1.1 Coho Salmon was 537 mm (n=15); age-2.1 was 539 mm (n=55); and age-3.1 was 535 mm (n=2).

The midpoint of the run occurred on August 25. The mid-50% of the run occurred from August 20 to August 30. Based on the historic median statistic, the Coho Salmon run was approximately 3 days later and 21.4% compressed than the historic median run.

In addition to the Fall Chum and Coho Salmon captured a total of two Pink Salmon *O. gorbushcha*, two Sockeye Salmon *O. nerka* and one sheefish *Stenodus nelma* were caught in the test fishery.

This project supported a resource management project in a cost-effective manner and facilitated communications between community and government entities. In addition, the project builds community capacity and is supported in the local area.

INTRODUCTION

The Yukon Area includes all waters of Alaska within the Yukon River drainage and coastal waters from Point Romanof, northeast of Kotlik, south to the Naskonat Peninsula. For management purposes, the Yukon Area is divided into 7 districts and 10 subdistricts (Figure 1). A description of the districts and subdistricts is provided in Estensen et al. (2022). Five species of Pacific salmon are found in the Yukon River drainage: Chinook salmon, *Oncorhynchus tshawytscha*, chum salmon, *O. keta*, Coho Salmon, *O. kisutch*, pink salmon, *O. gorbuscha*, and sockeye salmon, *O. nerka*, (Estensen et al. 2022). The chum salmon run is comprised of two genetically distinct races and runs, an early Summer Chum Salmon run, and a later Fall Chum Salmon run. Additional information regarding these two races of Yukon River chum salmon, along with the other four salmon species indigenous to the Yukon River drainage, can be found in Estensen et al. 2022. The 2022 Mountain Village Test Fish (MVTF) report documents test fishing activities for Fall Chum and Coho Salmon near the village of Mountain Village (RM 87) within the Lower Yukon Area (Figure 1). As in past years, the 2022 MVTF project began operation on July 18, 2022, and terminated operations on September 12, 2022.

Accurate and timely assessment of Yukon River Fall Chum and Coho Salmon run strength and timing as the runs pass through the lower Yukon River is a critical component of the information base used to determine harvestable surpluses of Yukon River salmon. Fall Chum Salmon runs have been relatively large through 2019 (Figure 2), with total run size averaging 898,400salmon for the most recent five-year period, 2017-2021, and averaging 995,800 for the most recent 10-year period, 2012-2021 (JTC 2022). However, recent total run sizes, starting in 2020, have been below the low end of the drainage-wide Biological Escapement Goal of 300,00 Fall Chum Salmon and have ranged from 95,000 in 2021 to 245,000 in 2022 (JTC 2023). The average run size for this 3-year period, 2020-2022, is 177,000 Fall Chum salmon.

During the period 2003-2019, concern over meeting subsistence needs and escapement requirements did not play a role in restricting subsistence and commercial fishing time during the fall fishing season. Substantial reductions in the Chinook salmon subsistence harvests during this period, because of the very poor Chinook salmon runs, may have caused subsistence fishermen to increase their subsistence harvest of Fall Chum and Coho Salmon. However, the corresponding loss of dog teams in the Yukon Area may have played a major role in substantially reducing the overall Fall Chum and Coho Salmon subsistence harvests over historic harvest levels. Since 2019, however, Fall Chum salmon runs have been very poor. Accordingly, commercial fisheries have not been allowed and subsistence fishing for all fish species has been either severely restricted or closed (JTC 2023).

Although the 2022 fall season salmon run sizes were improved over the record low 2021 run size, the Fall Chum and Coho salmon run sizes were considered poor. The 2022 Fall Chum Salmon run size of 242,000 (JTC 2023) salmon ranked 40 out of the 43 years of record since 1980 and the preliminary 2022 Coho Salmon of 110,000 (Bonnie Borba, ADF&G/CF Div Fairbanks, personal communication) fish, ranked 24 out of the 25 years of record since 1996 (Figure 2). Gear restrictions and closures on subsistence fishing throughout the fishing season negatively affected the subsistence harvest of all fish species (Figure 3; JTC 2023)

Based on survey and permit data, the 2022 preliminary subsistence salmon harvest in the Alaska portion of the Yukon River drainage was estimated to be 1,827 Chinook; 6,724 Summer Chum; 2,778 Fall Chum; and 1,090 Coho Salmon (Figure 3; JTC 2023). In 2022, Chinook, Summer Chum, Fall Chum, and Coho salmon harvests all fell well below their respective ranges of Amounts Reasonably Necessary for Subsistence (ANS) as defined by the Alaska Board of Fisheries (Brown and Jallen 2012). Additionally, the 2022 subsistence salmon harvests were all well below the most recent 5-year year average (2017-2021) subsistence harvest by species. Chinook Salmon was 94.0% below; Summer Chum Salmon subsistence harvest was 40.5% below; the Fall Chum Salmon subsistence harvest was 89.8% below; and the Coho Salmon subsistence harvest was 50.6% below the most recent average subsistence harvest.

Because of the restrictions on the commercial summer chum salmon gillnet fishery, and the incidental take of Chinook salmon in that fishery, more commercial emphasis had been placed on efficient management of the fall season runs. Now, with the complete closure of all commercial salmon fisheries and severe restrictions on the subsistence fisheries for all fish species, managers continue to strive to accurately assess inseason run sizes, in a timely manner, so that any harvestable surpluses can be quickly identified and more fully utilized. In addition to providing relative run strength, species composition, age-class composition, and run timing information, the fall season MVTF project serves as one of the checks among the three lower Yukon River assessment projects. This project either provides confidence in the information provided from the Lower Yukon Test Fish (LYTF) project and mainstem Yukon River sonar project, operated near the community of Pilot Station, or provides a basis for questioning data from any of these projects.

Fall Chum Salmon tend to be cyclic in run abundance (Figure 2). Therefore, most biologists expected that the large Fall Chum Salmon runs observed returning to the Yukon River from 2003 through 2019, would not persist. Prior to 2003, some Yukon River Fall Chum Salmon runs were so small that restrictions on commercial and, in some years, even subsistence harvest restrictions were implemented. In addition, because of the above-described implications of the low Chinook Salmon runs to the subsistence and commercial fisheries, more efficient management of the fall season salmon stocks is necessary, especially during low abundance years. Like current management strategies employed during the summer season because of the very low Chinook salmon runs, during times of low abundance of Fall Chum Salmon, managers must continue to accurately assess the run size in a timely manner, so that conservation efforts can be equitably distributed among all subsistence users within the Alaska portion of the drainage. Managers must also balance all Alaskan harvests with U.S./Canada Agreement obligations and escapement needs. The inseason and post-season information from the MVTF project plays a vital role in adding to the knowledge base of the Yukon River Fall Chum and Coho Salmon stocks. Therefore, it is imperative that this project continues during times of high abundance, as well as low abundance runs.

Project Justification:

The MVTF project is strategically located between two other important assessment projects that are separated by over 100 river miles and approximately three- and four-salmon travel days, depending on the salmon species and associated swimming speed. This project provides the first

composite assessment of the run after the stocks are assessed by the LYTF and merge into the mainstem from the three different mouths (Figure 1) within the delta area but before the fish are assessed at the mainstem Yukon River sonar project. In conjunction with the LYTF and mainstem Yukon River sonar projects, the fall season MVTF project will, in part, provide a comparative and often more accurate and timelier assessment of the inseason salmon run strengths and run timings as the salmon runs pass through the lower river. This project will also provide additional insight into the expected Fall Chum and Coho Salmon run strength at the mainstem Yukon River sonar site. Age, sex, size data from this project will be used in conjunction with the LYTF to provide insight into the characteristics of the test net catch, the subsistence harvest, and possibly the run. All fish retained will be distributed to village residents for subsistence purposes.

While salmon passage estimates from the mainstem Yukon River sonar project remains the key component of salmon run assessment, some years the sonar data has been affected by riverine conditions and riverbed morphology (Lozori and McIntosh. 2013). The river stage, debris load, and high turbidity affect the identification of sonar targets, as well as test fishing catches at the sonar site. It appears, however, that the MVTF project is generally not affected as much by river stage and debris to the extent as other Lower Yukon River projects. Therefore, the MVTF project may not only provide a check as to the accuracy of the mainstem Yukon River sonar counts attributed to Fall Chum and Coho Salmon but may also provide one of the most reliable timing assessment projects for Fall Chum Salmon in the lower Yukon River in some years. At least, in those years, differences will cause managers to pause and scrutinize data from the mainstem Yukon Sonar and LYTF projects but may also cause managers to weigh other information more heavily in their run assessment. In addition to comparisons among the lower river run assessment projects, age data will aid in the identification of trends in brood year assessment and may assist in future run forecasting.

Information from the MVTF project could also be used in conjunction with other project data, both inseason and post season, adjust or verify the species composition and run sizes of the salmon runs that were based on the test fishery in conjunction with the sonar project. Additionally, for the time in September after the sonar project terminates, the MVTF provides species-specific catch per unit effort (CPUE) estimates that are used to estimate the number of salmon that pass the sonar site, but are not counted by sonar, based on the inseason relationship between the projects when both projects were operational during the season.

Project Partnership:

The partnership among the principal investigator, Gene Sandone (G. Sandone Consulting, LLC, the Asacarsarmiut Tribal Council (ATC); Yukon Delta Fisheries Development Association (YDFDA); and ADF&G provides an opportunity for improving public support and participation in management of the fall season salmon runs. Expert fishermen from the village of Mountain Village, hired by ATC, are not only trained in proper test fishing techniques but are also encouraged to view their activities as being a contributing factor in the management of the fall season salmon runs.

I believe that participation in this project by ATC, YDFDA, and the expert fishermen hired to conduct the test fishery instills ownership in the management of the fishery because they see that

the data collected from this project is used in management decisions. On-site discussions and telephone conversations with the expert fishers and their helpers during the project regarding the management of the fisheries to achieve management objectives not only builds and maintains community capacity and support for the project but also encourages the understanding that there are additional upriver needs and demands on the same resource that passes their village. I believe that these conversations and discussions promote stewardship of the resource and result in a fuller understanding of the value of the salmon resource.

Project Goals:

The long-term project goal is to develop a CPUE and age, sex, and length (ASL) database that can be used, primarily and in conjunction with the mainstem Yukon River sonar abundance database to manage the fall season fisheries more efficiently. Further, this project seeks to build local capacity through participation of the local tribal government and expert fishers, foster understanding, and acceptance by project partners of the management of salmon resources, encourage local stewardship, and promote the value of the salmon resource throughout the drainage.

Objectives:

The specific objectives of this project are to:

1. Estimate the relative abundance (test fish CPUE) and run timing of the Yukon River Fall Chum and Coho Salmon run at Mountain Village.
2. describe the ASL composition of the Fall Chum and Coho Salmon caught in test drift nets; and
3. provide conservation and stewardship experience for rural residents.

Study Area:

The initial study area was upriver from the community of Mountain Village, Alaska. In the past, test fishing was conducted at three specifically identified locations approximately 5 miles upriver from the community of Mountain Village (Figure 4). Initial locations were identified as Site 1, Site 2, and Site 3 (Figure 4). Site 2 was located on the north side of a prominent sand bar in the general vicinity but southwest of Site 1. Prior to 2015, test fishing was conducted once a day, at noon at these three sites. However, because of the extremely low Fall Chum and Coho Salmon catches at Site 2 during 2014 and because the overall MVTF CPUE did not track well with the LYTF and mainstem Yukon River sonar run assessment during that year, it was decided, through consultations between ADF&G and Gene Sandone, to eliminate Site 2 from the test fishery, starting in 2015. Accordingly, starting in 2015, only two sites were fished, Sites 1 and 3 (Figure 4). Additionally, it was also decided that these two sites would be fished twice a day, at approximately 0800 and 1900 daily, to sample the salmon runs more completely and accurately. These two sites were fished twice a day, as described above, in this test fishery from 2016-2018 without any problems.

In 2019 changes in river morphology negatively affected Site 1, causing Site 1 to become inoperable. Additionally, a suitable alternative fishing site on the north bank of the Yukon River near Site 3 could not be established. Accordingly, the search for new fishing sites, downriver of the community of Mountain Village, was completed on July 23, 2019. These downriver fishing sites, Sites 5 and 6 (Figure 4) were established as the primary test fishing sites and were fished

on a twice-daily basis, starting on July 25, 2019, for the remainder of the season. After July 24, 2019, Site 1 was abandoned and not fished again. Site 3 was fished occasionally during the remainder of the 2019 season for comparative purposes. During 2019, all test fishing catches for all sites were used in the reporting of the daily catch as well as the calculation of the daily CPUE. Starting with the 2020 season, sites 5 and 6 (Figure 4) were exclusively fished on the twice-a-day schedule, originally established in 2015. These fishing sites and schedule were continued through the 2022 fall season.

METHODS

Test Fishing

G. Sandone Consulting, LLC, in cooperation with ATC and ADF&G, conducted a test fishery near the community of Mountain Village to monitor the Fall Chum and Coho Salmon runs and salmon characteristics relative to the gear used for capture. Test fishing was planned to commence on July 18 and continue for 57 days of scheduled test fishing through September 12. This period encompasses most of the entire fall season salmon migration and a substantial portion of the Coho Salmon migration. This schedule considers: 1) the approximate two-day lag time between the LYTF (RM 24), which begins fall season operation on July 16, and the MVTF location (RM 87) and 2) the approximate 1-day lag between the MVTF location and the mainstem Yukon River sonar project site (RM 124). Fall season operations started at the mainstem Yukon River sonar site on July 19. Accordingly, the first day of the Fall Chum Salmon migration passing Mountain Village is July 18. Note that a portion of the Fall Chum Salmon run passes the above-mentioned projects prior to the onset of the respective fall season operational date and a portion of the summer chum salmon run passes the projects after the start of the fall season at each respective project (Flannery and Wenburg 2015). However, it is beyond the scope of this study to adjust the CPUE and ASL information based on this overlap.

ADF&G staff traveled to Mountain Village on July 17 to train all the contract fishermen. Although ATC hired the individual fishermen, Gene Sandone of G. Sandone Consulting, LLC, managed the test fish crew, and was responsible for supervision and general oversight of the collection and timely reporting of the data. Residents were hired as professional fishermen by ATC and their expertise was employed in identifying drift sites. Contract fishermen and ATC were responsible for the submission of drift and biological data daily to ADF&G and to Gene Sandone. ATC was also responsible for the orderly distribution of the test fish catches to residents.

As in previous years, three contract fishermen were employed to test fish during the 2022 field season. One of the contract fishermen was also hired as a senior contract fisherman. This senior fisherman was hired to oversee the smooth and seamless transition from one contract fisherman to another and that the collection of all the data was consistent among all fishermen. The senior fisherman was also tasked with taking care of the nets during the off season and ensuring that all the nets were properly mended.

Test fishing with gillnets was conducted twice a day, separated by approximately 12 hours. Gillnet gear consists of a 25-fathom shackle of 5.875-inch stretch mesh, multi-filament drift gillnet. The net is 35 meshes deep and is constructed of mono-multifilament strands in a light brown or green color.

Two sets of data notebooks were provided to the test fishing crew for recording drift and biological data. One notebook was for recording daily test fishing information associated with the specific drifts. The second was for recording salmon sex and length (biological) information. These data notebooks were referred to as the drift data logbook and the biological data logbook, respectively. The inside page of the front cover of the drift notebook outlines the specific instructions for collecting data from the test fishing drifts, along with weather-related data. The inside of the back cover provides contact information. The inside page of the front cover of the biological data logbook outlines specific instructions for collecting sex and length data and scales from sampled Fall Chum and Coho Salmon. The inside of the back cover provides contact information and information on filling out the scale cards. In 2022, because of the severe restrictions imposed on the subsistence salmon fishery, a test fishing permit was included in both sets of data books.

Twice daily, during the first set of each drift period, wind speed and direction, air and water temperature, percent cloud cover, precipitation and river water conditions were noted and recorded. In the past, test fishing operations were not conducted when a commercial fishery opening was being prosecuted or were completed prior to the commercial fishing period. Because commercial fishing periods usually started in the afternoon and continued for 4 or 6 hours, only the evening test fishing drifts were affected. However, no commercial fishing periods were prosecuted during 2022.

Inseason, both the drift and biological data page photos were texted to ADF&G and Gene Sandone daily to be used in fishery management and provide quality control (QC). Both data notebooks and the collected fish scale cards were delivered to Gene Sandone after the MVTF project terminated for the season. After checking for accuracy, the scale cards were delivered to ADF&G for age determination.

Sampling/Statistical Design:

Salmon ASL sampling protocols developed and standardized by ADF&G staff (see Schumann and Dubois 2011) are used in sampling the salmon catch. CPUE calculations were based on the same method that ADF&G uses. Run timing assessment was based on ADF&G procedures. All methods and data analysis are based on ADF&G protocols. Additionally, in coordination with G.Sandone, Consulting LLC, ADF&G conducted the initial training of the fishermen in test fishing and sampling methods and procedures on July 17, one day before commencement of the fall test fishing season.

Objective 1: Estimate the relative abundance and run timing of the Fall Chum and Coho Salmon runs:

CPUE is the primary indication of relative run strength. Daily species-specific CPUE was

calculated as the weighted average of the salmon catch and the duration of the drift. Daily CPUE was based on the catch of each species, the amount of net fished, and mean fishing time. The species-specific CPUE index standardizes catch reporting as the number of Fall Chum or Coho Salmon caught in 100 fathoms of gear, expanded to one hour of fishing time. The daily CPUE for each species was calculated as follows;

$$\text{Weighted Daily CPUE}_d = \frac{(\text{total daily salmon catch}) \times 100 \text{ fm}}{((\text{net length}_1 \times \text{MFT}_1) \times (\text{net length}_2 \times \text{MFT}_2) \times (\text{net length}_3 \times \text{MFT}_3) \times (\text{net length}_4 \times \text{MFT}_4))} \quad (1)$$

where:

Daily salmon catch is species specific (Fall Chum or Coho Salmon),
MFT = mean fishing time for each set, in hours,
Net Length expressed in fathoms (fm),
CPUE_d is the daily CPUE weighted by MFT and net length employed.

Mean fishing time (*MFT*) is calculated as:

$$MFT = (C - B) + \frac{(B - A) + (D - C)}{2} \quad (2)$$

where:

A = time net deployment started,
B = time net fully deployed,
C = time net retrieval started, and
D = time net fully retrieved.

Fishing times were recorded to the nearest minute for each drift in the drift log data book. Time was recorded for the beginning of net deployment, when the net is fully deployed, when the net retrieval starts, and when the net is fully retrieved.

Each net was drifted for approximately 20 minutes unless catch rates were high. During high catch rates, the net soak time was subjectively assessed to be long enough to collect at least 30 salmon from each drift. Note that the net can capture fish prior to being fully deployed, and during the time it is being retrieved. Therefore, MFT for each set is adjusted by adding half of the summed total time to set and retrieve the net. However, when an estimated 10 salmon or more were observed in the net, the crew was instructed to pull the entire net into the boat, record the time, and then pick the salmon out of the net. The distance covered by the drift will depend on the time the net is in the water, as well as water currents and wind conditions.

On occasion and in general, test fishing will not be conducted because of rough water and unsafe boating conditions, test fishermen illness, mechanical issues, and the scheduling of a commercial fishing period coincidental with the test fish time. However, regardless of the number of drifts

conducted, the total catch along with the net length and MFT for each drift will be used in the calculation of the daily weighted CPUE statistic ($CPUE_d$) for developing relative abundance and timing information.

Missing daily CPUE values were estimated and provided by ADF&G. These estimates are calculated from relationships between the daily LYTF CPUE versus the MVTF CPUE and/or the MVTF CPUE versus the mainstem Yukon River sonar estimates.

Run timing of Fall Chum and Coho Salmon through the Lower Yukon Area will be assessed using the median day of passage along with the period when the mid-50% of the run passed the project. Quartile days will be defined based on the day when 25%, 50% and 75% of the run passed the project, based on the cumulative Fall Chum and Coho Salmon CPUE. The first and third quartile day defined the mid-50% of the run.

Weather data, fishing times, catch data, and other ancillary data regarding each drift were recorded in the MVTF drift log data book during the first morning and evening drifts.

Objective 2: Describe the ASL composition of the Fall Chum and Coho Salmon caught in test drift nets.

Sex was determined and recorded based on internal inspection of gonads. Length of each sampled Fall Chum and Coho Salmon was measured from mid-eye to fork of tail (METF), to the nearest 1 mm using a caliper. Sex and length data were recorded in the appropriate section of the biological data notebook. Mean length by sex and age was calculated along with 95% confidence intervals around the estimated mean length. One scale sample was collected from up to 30 Fall Chum Salmon and three scale samples from up to 20 Coho Salmon per day for subsequent age determination. Scales were taken from the left side of the fish, approximately two rows above the lateral line, on the diagonal from the posterior insertion of the dorsal fin to the anterior insertion of the anal fin (Koo 1955). This is known as the “preferred area”. Salmon scales taken from the preferred area were mounted on gum cards, or scale cards, in the field.

Information, including the species of salmon, location, statistical area, sampling date, gear, name of collector and any additional comments, was recorded on the back of each scale card. Fishermen were instructed to leave the card number blank. Each scale card had locations for 30 scales. They were numbered 1 through 30. Accordingly, each scale card could accommodate one scale from 30 Fall Chum salmon or three scales from 10 Coho Salmon. On days when more than 10 Coho salmon were sampled, two scale cards were used. On those days, each Coho Salmon scale card was uniquely identified with an “A” behind the date for the first scale card with the first 10 Coho Salmon scales and “B” behind the date for the second card with the remaining Coho Salmon scales. No more than one scale card was used to store scales from the daily maximum sampling target of 30 Fall Chum Salmon. Scale cards from each fisherman were stored with ATC prior to shipment. All scale cards were delivered to Gene Sandone for quality control assessment and the recording of the species-specific scale card number on the back of the card. Scale cards were subsequently delivered to ADF&G. ADF&G was responsible for processing and reading the scales for age determination. Salmon ages were expressed in European notation with the freshwater age followed by the saltwater age. All fish retained in the

MVTF were available for the residents to take for subsistence purposes once the catch was sampled.

Objective 3: Provide a conservation and stewardship experience for rural local residents.

This project through its local hire component and involvement of local tribal government provides an opportunity to build community capacity and stewardship. This project provides residents with work experience and training. Through discussions with the ADF&G and the project biologist, test fishermen achieved a sense of the importance of the data that they collected to the inseason management of the Yukon River Fall Chum and Coho Salmon runs. This project supports resource management in a cost-effective manner and facilitates communications between community and government entities. In addition, the project sought to build community capacity and is supported in the local area.

RESULTS

Test Fishing

During the 2022 season, test fishing was conducted from July 18 through September 12 inclusive, for a total of 57 possible days of test fishing days (Table 1). A total number of 168, 73.7%, out of a possible 228 test fishing drifts were successfully completed during the season (Table 1, 2; Appendix A1). Test fishing time at both sites varied considerably but combined for a total of 56.0 hours (Table 2). Out of the 57 possible days of test fishing and the four possible drifts per day, one drift was conducted on 3 days, 5.3%; two drifts were conducted on 19 days, 33.3%; three drifts were conducted on one day, 1.8%; and a full complement of four drifts were conducted on 31 days, 54.4%. Test fishing was not conducted, zero drifts, on 3 days, 5.3%, during the season because of hazardous river conditions or mechanical issues (Table 1, Appendix A1). Both the number of drifts conducted, and the overall test fishing duration was dissimilar at Site 5 and 6. The total number of drifts conducted at Site 5 was 103 drifts, 61.3%, and 65 drifts, 38.7% at Site 6. Likewise, the duration of all drifts at Site 5 totaled 34.3 hours, 61.3% and 21.7 hours, 38.7% at Site 6 (Table 2; Appendix A1). The difference between the number of drifts and the associated duration at the two sites was mainly attributed to the inability to successfully test fish at Site 6 because of high water conditions from the initiation of the 2022 season on July 18 through the morning drift on August 10. Test fishing at Site 6 was only consistently conducted after the morning drift on August 10 through September 12 (Table 1; Appendix A1).

Weather information was collected daily during the first morning and evening drift, at Site 5, except when test fishing at Site 5 was not conducted. During these times, as noted above, weather data were collected at the Mountain Village town site. Daily weather observations (Table 3) include instantaneous water and air temperature, occurrence and intensity of precipitation, cloud cover and fog, wind direction and velocity, and river water condition. Daily instantaneous water temperature data collected in conjunction with the MVTF project, along

with the daily mean water temperature and mean discharge observed at the Pilot Station USGS gauging station (http://waterdata.usgs.gov/usa/nwis/uv?site_no=15565447) are presented in Figure 5. Daily instantaneous morning (AM) and evening (PM) air temperatures associated with the MVTF project are presented in Figure 6. Since water discharge can affect fish catchability, the 2022 data are compared to discharge measurements for the Yukon River at the Pilot Station USGS gauging station from 2014 (Figure 7). Associated mean water temperature data observed on the Yukon River at the Pilot Station USGS gauging station for 2014-2021 are presented in Figure 8.

Objective 1: Estimate the relative abundance and run timing of the Fall Chum and Coho Salmon runs:

Fall Chum Salmon

A total of 155 Fall Chum Salmon were caught during test fishing operations associated with the MVTF in 2022. This total includes 10 fish identified as Fall Chum Salmon that dropped out of the net as it was being retrieved and three Fall Chum Salmon caught on an aborted drift on August 9th at Site 6, drift # 92 (Appendix A1). During this drift, the net was caught on a snag and was left in the water overnight. In consultation with ADF&G, it was decided that the three Fall Chum Salmon caught in Drift 92 would not be included in the CPUE analysis because of the inordinate time the net was fishing and because the net was not fishing properly. However, although not included in the CPUE analysis, these three salmon were sampled for Age, Sex, and Size (ASL) and included in the ASL analysis. Therefore, the total number of Fall Chum Salmon used in the calculation of the daily CPUE was 152 (Appendix A 1; Table 1 and 2).

Fall Chum Salmon catches were very dissimilar at the two test fishery sites. A Total of 49 Fall Chum salmon, 32.2%, were caught at Site 5 in 103 drifts and 103 Fall Chum Salmon, 67.8%, were caught at Site 6 in 65 drifts (Table 2). Note that although the number of drifts and time spent drifting, or duration, at Site 5 was 61.3% of the total, the Fall Chum Salmon catch was only 32% of the total. The Fall Chum Salmon catch per hour for Site 5 was 1.43 Fall Chum Salmon, while the catch rate for Site 6 was 1.84, 28.7% higher than the catch rate of site 5.

The total Fall Chum Salmon MVTF catch in 2022, 152 salmon, was 43.2% below than the historic (1997-2012, 2014-2021) median catch of 267.5 Fall Chum Salmon (Arctic, Yukon, Kuskokwim Data Management System (AYKDMS)). Fall Chum Salmon catch ranged from zero, on 15, 29.4%, of the 51 days when at least two test fishing drifts were successfully prosecuted, to 18 Fall Chum Salmon on August 17 (Table 1). Likewise, the daily CPUE ranged from zero on the 18 days to a daily high of 108.00, likewise on August 17 (Table 1; Figure 9). On this day, only the two morning drifts were conducted (Appendix A1). Interestingly, all 18 fish were captured in one drift, drift # 122, at Site 6 (Appendix A1).

The 2022 MVTF total cumulative CPUE for Fall Chum Salmon during the period, July 18 through September 12, 2021, totaled 607.00. (Table 1). The 2022 cumulative CPUE was 70.2% below the historic median (1997–2022 excluding 2013) cumulative CPUE of 2,035.33 (Figure 10; AYKDMS). However, this comparison is confounded by several reasons, including the varying number of test drifts per day, varying number of test fishing sites employed; the various

location of the test fishing sites; varying efficiencies of the fishing sites; the varying expertise of the fishers; the varying duration of the project over the years; and the variable environmental conditions and river morphology.

The mid-50% of the 2022 Fall Chum Salmon run at the MVTF site occurred from August 10 through August 23 and encompassed 14 days. The midpoint of the 2022 run occurred on August 17 (Table 1; Figures 9 and 10). The 2022 Fall Chum Salmon run median day of passage occurred 8 days later than the historic median day of passage, August 9 (Figures 9 and 10: AYKDMS). The 2022 first quartile day of passage, August 10, occurred 12 days later than the historic first quartile day of passage, July 29. The 2022 third quartile day of passage, August 23, occurred five days later than the historic third quartile day of passage, August 18 (Figures 9 and 10). The duration of the median historic (1997–2019 excluding 2013) mid-50% of the run occurred from July 29 through August 18 and encompassed 21 days. The 2022 Fall Chum Salmon run was approximately 8 days later, based on the median day of passage, and 33.3% more compressed than the historic median Fall Chum Salmon run (Figures 9 and 10).

Coho Salmon

A total of 83 Coho Salmon were caught during test fishing operations associated with the MVTF in 2022. This total includes three 3 fish identified as Coho Salmon that dropped out of the net as it was being retrieved and one Coho Salmon caught on an aborted drift on August 9th at Site 6, drift # 92 (Appendix A1). During this drift, the net was caught on a snag and was left in the water overnight. In consultation with ADF&G, it was decided that the one Coho Salmon caught in Drift 92 would not be included in the CPUE analysis because of the inordinate time the net was fishing and because the net was not fishing properly. However, although not included in the CPUE analysis, this one salmon was sampled for Age, Sex, and Size (ASL) and included in the ASL analysis. Therefore, the total number of Coho Salmon used in the calculation of the daily CPUE was 82 (Appendix A 1; Table 1 and 2).

Coho Salmon catches were very dissimilar at the two test fishery sites. A Total of 26 Coho Salmon, 31.7%, were caught at Site 5 in 103 drifts and 56 Coho Salmon, 68.3%, were caught at Site 6 in 65 drifts (Table 2). Note that although the number of drifts and time spent drifting, or duration, at Site 5 was 61.3% of the total, the Coho Salmon catch was only 31.7% of the total Coho Salmon catch. The catch per hour for Site 5 was 0.76 Coho Salmon, while the catch rate for Site 6 was 2.58, 3.39 times the catch rate of site 5 (Table 2; Appendix A1).

The 2022 Coho Salmon MVTF catch used the CPUE calculation in 2022, 82 salmon (Table 1 and Table 2), was 60.7% below than the historic (1997-2012, 2014-2021) median catch of 208.5 Coho Salmon (AYKDMS; Table 1 and 2). Coho Salmon catch ranged from zero, on 23 days, or on 45.1%, of the 51 days when at least two drifts were successfully prosecuted to 9 Coho Salmon caught on August 27 (Table 1). Likewise, the daily CPUE ranged from zero to a daily high of 27.00 on August 27 (Table 1; Figure 11).

The 2022 total cumulative MVTF CPUE for Coho Salmon during the period, July 18 through September 12, 2022, totaled 259.00 (Table 1, Figure 14). The 2022 cumulative CPUE is 76.5% below the historic median cumulative CPUE of 1,101.39 (Figure 12; AYKDMS). However, like Fall Chum Salmon, this comparison is confounded by several reasons, including the varying

number of drifts per day, the varying number of test fishing sites employed; the various location of the test fishing sites; varying efficiencies of the fishing sites; the varying expertise of the fishers; the varying duration of the project over the years; and the variable environmental conditions and river morphology.

The mid-50% of the 2022 Coho Salmon run at the MVTF site occurred from August 20 through August 30 and encompassed 11 days. The midpoint of the 2022 Coho Salmon run occurred on August 25 (Table 1; Figures 11 and 12). The first quartile day of passage of the 2022 Coho Salmon run was three days later than the historic first quartile day of passage; the 2022 median day of passage was also three days later than the historic median day of passage; and the 2022 third quartile day of passage occurred on the historic third quartile day of passage. The 2022 Coho Salmon run was late by three days and the mid-50% of the run was 21.4% compressed over the median historic mid-50% passage (Figures 11 and 12).

Other Species

In addition to the Fall Chum and Coho Salmon captured a total of two Pink Salmon, two Sockeye Salmon and one sheefish, *Stenodus nelma*, were caught in the test fishery. All fish were retained.

Objective 2: Describe the ASL composition of the Fall Chum and Coho Salmon caught in test drift nets.

ASL records of individual Fall Chum and Coho Salmon are archived in the AYKDMS (<http://www.adfg.alaska.gov/CommFishR3/WebSite/AYKDBMSWebsite/Default.aspx>). Therefore, the individual records of sampled fish are not included in this report. However, summaries are provided.

Fall Chum Salmon

A total of 145 Fall Chum Salmon were sampled for ASL during the 2022 MVTF operations (Table 4). A total of 137, or 94.5%, were successfully aged. All 145 Fall Chum Salmon sampled were successfully sexed and measured (Table 4).

Female Fall Chum Salmon comprised 57.7% of the aged sample, 75.0% of the unaged sample, and 58.6% of the total sample (Table 4). There was no discernable trend in female composition throughout the run. Female composition ranged from 46.3% in the second quartile to 68.0% observed in the first quartile. Percentage of females in the remaining quartiles exceeded 60% (Table 5).

Age-0.3 dominated the aged sample, accounting for 74.5% of the aged sample, followed by age-0.4 salmon, 24.1%, and age-0.2 salmon at 1.5% (Tables 4 and 5). No age 0.5 were sampled. Age-0.3 salmon dominated each quartile, ranging from 62.2% in the second quartile to 91.7% in the fourth quartile. There was no trend in the age class composition over time (Table 5).

Mean length of Fall Chum Salmon data by age, sex, aged, unaged, and combined are presented in Table 4. Mean length of all sampled Fall Chum Salmon was 563 mm (n=145, Table 4). Mean length for male Fall Chum Salmon was 564 mm (n=60). Mean length for female Fall Chum Salmon was 562 mm (n=85). Mean length for age-0.2 Fall Chum Salmon was 510 mm (n=2); 561 mm for age-0.3 (n=102); and 571 mm for age-0.4 Fall Chum Salmon (n=33; Table 4). Inspection of the 95% confidence intervals (CI) around the mean length of sampled Fall Chum Salmon (Figure 13) indicates that age-0.2 female salmon were significantly smaller than both male and female age-0.3 and age-0.4 salmon, all aged salmon, male and combined unaged salmon, and all salmon combined (Figure 13). Otherwise, there may be no other significant differences among all age groups, sex groups, or between aged and unaged sample populations. (Figure 13).

Coho Salmon

A total of 80 Coho Salmon were sampled for ASL during the 2022 MVTF operations (Table 6; AYKDMS). Of this total, 72 fish, or 90.0%, were successfully aged. All sampled Coho salmon were successfully sexed and measured (Table 6),

Female Coho Salmon comprised 38.9% (n=28) of the aged sample, 62.5% (n=5) of the unaged sample, and 41.3% (n=33) of the total sample (Table 6). Percent female increased from 34.8% in the first quartile to 42.9% and 52.4% in the second and third quartile, respectively, but decreased to 36.4% in the fourth quartile. Likely, because of the small sample size, a more pronounced increase of female salmon throughout the duration of the total run, especially in the fourth quartile, was not detected (Table 7).

Age-2.1 dominated the sample, accounting for 76.4% (n=55) of the aged sample, followed by age-1.1 at 20.8% (n=15), and age 1.3 at 2.8% (n=2) (Table 6 and Table 7). The mean length of all Coho Salmon measured was 538 mm (n=80). Mean length of male Coho Salmon was 539 mm (n=47, Table 6). Mean length of female Coho Salmon was 537 mm (n=33; Table 6). Mean length for age-1.1 Coho Salmon was 537 mm (n=15), 539 mm for age-2.1 salmon (n=55), and 535 (n=2) for age 3.1 salmon (Table 6). Inspection of the 95% confidence intervals (CI) around the mean length of sampled Coho Salmon (Figure 14) indicates that there may be no significant differences among all age groups, sex groups, or between aged and unaged sample populations.

Objective 3: Provide a conservation and stewardship experience for rural local residents.

The partnership among the principal investigator, Gene Sandone (G. Sandone Consulting, LLC), ATC; YDFDA; and ADF&G provided the opportunity for improving public support and participation in management of the fall season salmon runs. Expert fishers from the village of Mountain Village were hired by ATC and were not only trained in proper test fishing techniques but are also encouraged to view their activities as being a contributing factor in the management of the fall season salmon runs. Summary data information sheets were provided by ADF&G to ATC on a weekly basis.

I believe that participation in this project by ATC, YDFDA, and the expert fishermen hired to conduct the test fishery instills ownership in the management of the fishery because they see that the data collected from this project was used in management decisions. On-site discussions with the expert fishermen and their helpers, either during the pre-season training period or in conversations during the project, regarding the management of the fisheries to achieve management objectives, the U.S./Canada Agreement and the U.S. Agreement obligations to Canada, and the upriver commercial and subsistence fisheries not only builds and maintains community capacity and support for the project but also encourages the understanding that there are additional upriver needs and demands on the same resource that passes by their village. I believe that these conversations and discussions promote stewardship of the resource and result in a more fully understanding of the value of the salmon resource.

Future MVTF Site Locations

Substantial decreases in the salmon catch and associated CPUE at Site 1 prompted the abandonment of Site 1 in 2019. Because an alternative site on the North Bank of Yukon River near Site 3 could not be found, the two test fishing sites had to be relocated. This prompted the relocation of the test fishing sites to sites downriver of the community of Mountain Village (Figure 4) during the 2019 season. Accordingly, sites 5 and 6 were established in 2019. The MVTF operations continued to use these new test fishing locations for the 2022 season. However, during the 2022 season, Site 6 (Figure 4) was not consistently fished until the evening drift on August 10 because the prominent sand bar associated with the site was underwater and the water at the site was too deep to fish effectively. Prior to, or during the 2023 fishing season, G.Sandone Consulting, along with ADF&G, ATC, and the expert fishermen will decide whether to fish at Site 5 and 6 or return to the upriver historic sites, Site 1 and 3 (Figure 4). High water events during the past couple years may have scoured Site 1 and changed the river morphology that would allow the salmon to return to that part of the river. Accordingly, if this did occur, we would probably return to fish those historic sites.

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TABLES

Table 1. Total daily catch and associated Catch Per Unit Effort for Fall Chum and Coho Salmon, Mountain Village Test Fish project, July 18–September 12, 2022.

		Fall Chum Salmon				Coho Salmon					
Date (2022)	Number of Drifts ^a	Catch ^b	Avg. Daily CPUE	Cum CPUE	prop ^c	Catch ^b	Avg. Daily CPUE	Cum CPUE	prop ^c	Other fish ^d	
18-Jul	2 ^e	0	0.00	0.00	0.000	0	0.00	0.00	0.000		
19-Jul	2 ^e	0	0.00	0.00	0.000	0	0.00	0.00	0.000		
20-Jul	1 ^{e,f}	1	12.00	12.00	0.020	0	0.00	0.00	0.000		
21-Jul	0 ^g	0	18.00 ^h	30.00	0.049	0	0.00	0.00	0.000		
22-Jul	2 ^e	4	24.00	54.00	0.089	0	0.00	0.00	0.000		
23-Jul	2 ^e	2	12.00	66.00	0.109	0	0.00	0.00	0.000		
24-Jul	2 ^e	0	0.00	66.00	0.109	0	0.00	0.00	0.000	1 PS	
25-Jul	0 ^g	0	4.00 ^h	70.00	0.115	0	0.00	0.00	0.000		
26-Jul	0 ^g	0	8.00 ^h	78.00	0.129	0	0.00	0.00	0.000		
27-Jul	1 ^{j e}	1	12.00	90.00	0.148	0	0.00	0.00	0.000		
28-Jul	2 ^e	0	0.00	90.00	0.148	0	0.00	0.00	0.000		
29-Jul	2 ^e	1	6.00	96.00	0.158	0	0.00	0.00	0.000		
30-Jul	2 ^e	1	6.00	102.00	0.168	0	0.00	0.00	0.000		
31-Jul	2 ^e	0	0.00	102.00	0.168	0	0.00	0.00	0.000		
1-Aug	2 ^e	0	0.00	102.00	0.168	0	0.00	0.00	0.000		
2-Aug	2 ^e	0	0.00	102.00	0.168	0	0.00	0.00	0.000		
3-Aug	2 ^e	0	0.00	102.00	0.168	0	0.00	0.00	0.000	1 PS	
4-Aug	2 ^e	0	0.00	102.00	0.168	0	0.00	0.00	0.000		
5-Aug	2 ^e	1	6.00	108.00	0.178	0	0.00	0.00	0.000		
6-Aug	2 ^e	0	0.00	108.00	0.178	0	0.00	0.00	0.000		
7-Aug	2 ^e	1	6.00	114.00	0.188	0	0.00	0.00	0.000		
8-Aug	1 ^{e k}	0	0.00	114.00	0.188	0	0.00	0.00	0.000		
9-Aug	2 ^l	4	24.00	138.00	0.227	0	0.00	0.00	0.000		
10-Aug	3 ^m	7	28.00	166.00	0.273	1	4.00	4.00	0.015		
11-Aug	4	2	6.00	172.00	0.283	0	0.00	4.00	0.015		
12-Aug	4	2	6.00	178.00	0.293	1	3.00	7.00	0.027		

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Table 1. Page 2 of 3.

		Fall Chum Salmon				Coho Salmon					
Date (2022)	Number of Drifts ^a	Catch ^b	Avg. Daily CPUE	Cum CPUE	prop ^c	Catch ^b	Avg. Daily CPUE	Cum CPUE	prop ^c	Other fish ^d	
13-Aug	4	8	24.00	202.00	0.333	0	0.00	7.00	0.027		
14-Aug	4	6	18.00	220.00	0.362	1	3.00	10.00	0.039		
15-Aug	4	2	6.00	226.00	0.372	1	3.00	13.00	0.050		
16-Aug	4	6	18.00	244.00	0.402	2	6.00	19.00	0.073		
17-Aug	2 ⁿ	18	108.00	352.00	0.580	2	12.00	31.00	0.120		
18-Aug	4	14	42.00	394.00	0.649	3	9.00	40.00	0.154		
19-Aug	4	8	24.00	418.00	0.689	6	18.00	58.00	0.224		1 RS
20-Aug	4	7	21.00	439.00	0.723	5	15.00	73.00	0.282		
21-Aug	4	4	12.00	451.00	0.743	1	3.00	76.00	0.293		
22-Aug	4	1	3.00	454.00	0.748	3	9.00	85.00	0.328		1 SF
23-Aug	4	2	6.00	460.00	0.758	6	18.00	103.00	0.398		
24-Aug	4	5	15.00	475.00	0.783	5	15.00	118.00	0.456		
25-Aug	2 ^f	0	0.00	475.00	0.783	2	12.00	130.00	0.502		
26-Aug	4	0	0.00	475.00	0.783	2	6.00	136.00	0.525		1 RS
27-Aug	4	5	15.00	490.00	0.807	9	27.00	163.00	0.629		
28-Aug	4	7	21.00	511.00	0.842	7	21.00	184.00	0.710		
29-Aug	4	1	3.00	514.00	0.847	3	9.00	193.00	0.745		
30-Aug	4	2	6.00	520.00	0.857	1	3.00	196.00	0.757		
31-Aug	4	13	39.00	559.00	0.921	2	6.00	202.00	0.780		
1-Sep	4	4	12.00	571.00	0.941	3	9.00	211.00	0.815		
2-Sep	4	1	3.00	574.00	0.946	3	9.00	220.00	0.849		
3-Sep	4	0	0.00	574.00	0.946	0	0.00	220.00	0.849		
4-Sep	4	0	0.00	574.00	0.946	1	3.00	223.00	0.861		
5-Sep	4	3	9.00	583.00	0.960	0	0.00	223.00	0.861		
6-Sep	4	5	15.00	598.00	0.985	4	12.00	235.00	0.907		
7-Sep	4	1	3.00	601.00	0.990	0	0.00	235.00	0.907		

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Table 1. Page 3 of 3.

		Fall Chum Salmon				Coho Salmon					
Date (2022)	Number of Drifts ^a	Catch ^b	Avg. Daily CPUE	Cum CPUE	prop ^c	Catch ^b	Avg. Daily CPUE	Cum CPUE	prop ^c	Other fish ^d	
8-Sep	4	1	3.00	604.00	0.995	0	0.00	235.00	0.907		
9-Sep	4	0	0.00	604.00	0.995	2	6.00	241.00	0.931		
10-Sep	4	0	0.00	604.00	0.995	1	3.00	244.00	0.942		
11-Sep	4	1	3.00	607.00	1.000	5	15.00	259.00	1.000		
12-Sep	4	0	0.00	607.00	1.000	0	0.00	259.00	1.000		
Totals	168	152	607.00			82	259.00				

^a Project sites since 2015 were altered from historical by fishing sites 1 and 3 two times per day (~12 h apart) instead of sites 1, 2, and 3 once a day at noon. During 2019, Sites 1 was abandoned; Site 3 was intermittantly fished during the season and 2 primary test fishing sites, Sites 5 and 6, were located on July 23 and fished throughout the season starting on July 25.

^b Includes retained, released, and fish that dropped out of the net.

^c The shaded box indicates the middle 50% of the cumulative CPUE. The median day of the cumulative CPUE is indicated by the box within the larger box.

^d Other fish include: PS=Pink Salmon, KS=Chinook Salmon; RS= Sockeye Salmon; Sf=sheefish; WF=white fish.

^e Site 6 was not fished because of extremely high water covering the sand bar.

^f Morning test fishing drift was cancelled because of hazardous river water conditions.

^g Test fishing drifts were cancelled because of hazardous river water conditions.

^h ADF&G estimated CPUE.

^j Morning test fishing drift was cancelled because of boat mechanical issues.

^k Evening test fishing drift were cancelled because of hazardous river water conditions.

^l Test fishing initiated at Site 6 but the net was caught on a snag and remained overnight. Because of this, the 3 Fall Chum and 1 Coho Salmon that were caught in this drift were excluded from the total catch and CPUE calculations. These fish were sampled and included in the ASL data and analyses. Data from this drift included in any other analysis.

^m Site 6 was not fished during the morning drift because of high water over the sand bar.

Table 2. Summary table of the total number of drifts, fishing duration, and Fall Chum and Coho Salmon catches at the Mountain Village Test Fish project test fishing sites fished during the 2022 MVTF season.

Site Number	Test Fish Drifts				Test Fish Catch							
	Number		Duration		Fall Chum Salmon				Coho Salmon			
					Retained	Dropout	Total Catch		Retained	Dropout	Total Catch	
	n	%	Hours	%	n	n	n	%	n	n	n	%
5 ^a	103	61.3%	34.3	61.3%	46	3	49	32.2%	25	1	26	31.7%
6 ^{b,c}	65	38.7%	21.7	38.7%	96	7	103	67.8%	54	2	56	68.3%
Total	168	100.0%	56.0	100.0%	142	10	152	100.0%	79	3	82	100.0%

^a Test fishing conducted during the entire season, July 18-September 12, 2023.

^b Test fishing was delayed until the evening drift on August 10, 2023 because of high water.

^c Does not include 3 Fall Chum and 1 Coho salmon caught on August 9 in Drift #92. The net was caught on a snag and remained overnight. Therefore, these fish were not included in the CPUE calculations but were sampled and included in the ASL analyses.

Table 3. Weather observations collected at Site 1 of the Mountain Village Test Fish project, July 18–September 12, 2022.

Date	Time	Temperature		Sky ^a	Wind		Precip. ^c	Water Cond.
		Air (°C)	Water (°C)		Dir.	Speed ^b (mph)		
18-Jul	am	6.5	17.8	4	S	0-5	N	sm waves
18-Jul	pm	12.0	18.8	2	W	16-20	N	choppy
19-Jul	am	6.1	16.3	3	W	6-10	N	sl chop
19-Jul	pm	11.0	17.1	5	SW	11-15	I	rough
20-Jul	am	8.8	na	na	na	na	na	na
20-Jul	pm	10.0	16.5	5	SE	16-20	I	choppy
21-Jul	am	6.3	na	5	W	26-30	N	hazard
21-Jul	pm	9.1	na	5	SW	26-30	I	hazard
22-Jul	am	7.4	15.3	5	SW	5-10	C	sl chop
22-Jul	pm	10.0	14.5	5	W	5-10	I	sl chop
23-Jul	am	8.7	13.9	5	SE	0-5	I	sm waves
23-Jul	pm	9.9	15.3	5	SE	0-5	I	sm waves
24-Jul	am	8.9	14.6	2	S	0-5	N	sm waves
24-Jul	pm	14.9	14.6	4	SE	0-5	N	sm waves
25-Jul	am	na	na	5	SE	21-25	I	rough
25-Jul	pm	na	na	5	W	26-30	I	hazard
26-Jul	am	9.3	15.2	5	SW	15-20	I	choppy
26-Jul	pm	8.5	na	5	SW	21-25	I	rough
27-Jul	am	8.5	na	6	SW	0-5	N	sm waves
27-Jul	pm	13.0	13.9	5	SW	0-5	N	sm waves
28-Jul	am	na	na	1	SW	0-5	N	sm waves
28-Jul	pm	18.2	14.1	5	NE	0-5	N	sm waves
29-Jul	am	10.6	na	5	E	5-10	N	sm waves
29-Jul	pm	20.0	14.4	5	E	5-10	N	sl chop
30-Jul	am	13.0	14.0	4	E	6-10	N	sl chop
30-Jul	pm	19.6	15.0	5	E	6-10	I	sm waves
31-Jul	am	13.7	14.8	5	E	0-5	N	sm waves
31-Jul	pm	13.4	15.0	5	E	11-15	I	sl chop
1-Aug	am	11.1	14.6	5	E	0-5	N	sm waves
1-Aug	pm	14.4	14.8	5	S	0-5	N	sm waves
2-Aug	am	11.1	14.0	5	W	6-10	C	sl chop
2-Aug	pm	14.2	14.8	5	W	6-10	N	sl chop
-continued-								

Table 3. Page 2 of 4.

Date	Time	Temperature		Sky ^a	Wind		Precip. ^c	Water Cond.
		Air (°C)	Water (°C)		Dir.	Speed ^b (mph)		
3-Aug	am	10.4	14.7	5	W	0-5	N	sm waves
3-Aug	pm	14.7	15.2	4	E	6-10	N	sl chop
4-Aug	am	9.4	14.9	5	S	6-10	I	sl chop
4-Aug	pm	11.5	14.8	5	E	11-15	I	sl chop
5-Aug	am	9.9	14.6	5	W	0-5	I	sm waves
5-Aug	pm	12.2	14.4	5	SW	6-10	C	sl chop
6-Aug	am	7.7	13.8	3	NW	11-15	N	choppy
6-Aug	pm	12.0	14.5	4	W	11-15	N	rough
7-Aug	am	10.7	14.2	5	SE	16-20	C	rough
7-Aug	pm	11.1	14.4	4	SW	11-15	I	rough
8-Aug	am	6.0	13.9	4	W	11-15	I	choppy
8-Aug	pm	8.3	13.7	4	SW	16-20	N	very rough
9-Aug	am	9.4	13.8	5	NW	6-10	I	sm waves
9-Aug	pm	12.2	14.3	4	NW	11-15	N	sl chop
10-Aug	am	7.3	13.9	5	SW	0-5	N	sm waves
10-Aug	pm	14.3	14.5	2	NW	6-10	N	sm waves
11-Aug	am	11.2	14.1	2	NW	0-5	N	sm waves
11-Aug	pm	15.8	14.7	2	NW	6-10	N	sm waves
12-Aug	am	9.6	14.2	5	NW	6-10	N	sl chop
12-Aug	pm	12.0	14.4	5	NW	6-10	N	sm waves
13-Aug	am	10.0	14.1	5	NW	6-10	N	sm waves
13-Aug	pm	11.9	14.4	2	NW	6-10	I	sl chop
14-Aug	am	8.3	13.3	5	NW	11-15	I	rough
14-Aug	pm	9.6	14.0	5	NW	16-20	I	rough
15-Aug	am	8.1	13.6	5	NW	6-10	N	sl chop
15-Aug	pm	8.9	13.8	5	S	6-10	I	sm waves
16-Aug	am	5.7	17.7	4	N	11-15	N	sm waves
16-Aug	pm	10.0	13.9	5	N	16-20	N	very rough
17-Aug	am	8.9	13.6	5	NW	11-15	C	very rough
17-Aug	pm	9.1	13.5	4	NW	11-15	I	hazard
18-Aug	am	6.1	13.1	5	SW	6-10	I	sl chop
18-Aug	pm	9.2	13.7	5	S	6-10	N	sm waves
-continued-								

Table 3. Page 3 of 4.

Date	Time	Temperature		Sky ^a	Wind		Precip. ^c	Water Cond.
		Air (°C)	Water (°C)		Dir.	Speed ^b (mph)		
19-Aug	am	5.2	13.2	4	NE	11-15	N	sl chop
19-Aug	pm	12.5	15.5	4	NE	11-15	I	sl chop
20-Aug	am	10.3	13.4	5	NE	6-10	N	sm waves
20-Aug	pm	13.2	13.6	5	NE	11-15	I	sl chop
21-Aug	am	11.7	13.4	5	NE	6-10	N	sm waves
21-Aug	pm	14.7	14.8	4	NE	6-10	I	sl chop
22-Aug	am	12.0	13.5	4	N	6-10	N	sm waves
22-Aug	pm	13.5	13.9	3	N	11-15	N	sm waves
23-Aug	am	9.8	13.6	5	NW	6-10	I	sm waves
23-Aug	pm	13.3	13.9	4	SW	0-5	N	sm waves
24-Aug	am	12.1	13.7	4	S	0-5	I	calm
24-Aug	pm	16.1	13.9	4	S	11-15	N	choppy
25-Aug	am	10.8	13.4	5	SE	26-30	C	hazard
25-Aug	pm	11.4	13.5	5	SW	16-20	I	rough
26-Aug	am	10.4	13.2	5	SW	6-10	C	sm waves
26-Aug	pm	11.2	13.6	5	SW	0-5	I	calm
27-Aug	am	10.9	13.5	5	SE	6-10	I	sm waves
27-Aug	pm	12.9	14.2	4	NE	0-5	I	calm
28-Aug	am	10.7	13.6	6	NW	11-15	I	sm waves
28-Aug	pm	12.8	13.9	4	NW	11-15	N	sl chop
29-Aug	am	6.6	13.1	4	NW	6-10	N	sm waves
29-Aug	pm	12.1	13.7	4	NW	0-6	N	calm
30-Aug	am	9.5	13.3	5	SE	11-15	C	sl chop
30-Aug	pm	12.1	13.5	5	SE	16-20	C	sl chop
31-Aug	am	10.7	13.1	5	S	6-10	C	sm waves
31-Aug	pm	11.0	13.2	5	S	0-5	C	calm
1-Sep	am	9.1	12.3	5	S	6-10	C	sm waves
1-Sep	pm	10.0	13.1	5	S	0-5	I	calm
2-Sep	am	7.7	11.3	4	N	0-5	N	calm
2-Sep	pm	14.0	13.2	3	NW	6-10	N	calm
3-Sep	am	9.7	12.3	5	NW	11-15	N	sl chop
3-Sep	pm	10.9	12.7	3	NW	11-15	N	sl chop
-continued-								

Table 3. Page 4 of 4.

Date	Time	Temperature		Sky ^a	Wind		Precip. ^c	Water Cond.
		Air (°C)	Water (°C)		Dir.	Speed ^b (mph)		
4-Sep	am	8.0	11.8	6	SW	0-5	I	calm
4-Sep	pm	10.4	12.4	5	SW	0-5	I	calm
5-Sep	am	8.0	12.0	5	SE	6-10	I	sm waves
5-Sep	pm	9.2	12.1	5	S	11-15	I	sl chop
6-Sep	am	8.0	11.9	5	SE	6-10	I	sm waves
6-Sep	pm	10.0	12.3	4	S	0-6	N	calm
7-Sep	am	6.4	11.4	6	E	0-6	N	calm
7-Sep	pm	10.3	12.1	4	SE	11-15	I	sl chop
8-Sep	am	6.3	11.5	3	NE	6-10	N	calm
8-Sep	pm	10.0	12.1	5	NW	16-20	I	rough
9-Sep	am	8.7	11.5	5	NW	11-16	I	choppy
9-Sep	pm	10.0	11.7	5	NW	16-20	N	rough
10-Sep	am	6.2	11.2	4	S	0-5	N	calm
10-Sep	pm	11.9	11.8	4	E	11-18	N	choppy
11-Sep	am	10.0	11.3	5	S	6-10	I	sm waves
11-Sep	pm	11.0	11.8	5	SE	16-20	C	choppy
12-Sep	am	10.0	11.3	5	E	16-20	I	sl chop
12-Sep	pm	12.0	11.8	5	E	16-20	I	choppy

^a Time: am or pm. Weather condition data collected during the first morning (am) first evening (pm) test drift while the test net was soaking.

^b Sky codes: 1=Clear and visibility unlimited; 2= cloud cover ≤ 25%;
3=Cloud cover ≤ 50%; 4=Cloud cover > 50%; 5=Complete overcast;
6= thick fog.

^c Wind speed was estimated based on individual test fisher assessment.

^d Precipitation codes: N=None, I=Intermittent, C=Continuous.

^e Water Conditions; sm waves=small waves; sl chop=slightly choppy;
hazard = hazardous conditions.

^f Test fishing not conducted because of hazardous river water conditions
Weather data were collected from Mountain Village.

^g Test fishing not conducted at Site 6 because of hazardous river water conditions.

Table 4. Fall Chum Salmon age, sex, and size summary, by brood year, for salmon caught and sampled in association with the Mountain Village Test Fish project, July 18 – September 12, 2022.

Brood Year (Age)														
	2019 (0.2)		2018 (0.3)		2017 (0.4)		2016 (0.5)		Aged Sample ^{a,d}		Unaged sample ^b		Total sampled ^{c,d}	
NUMBER SAMPLED														
	n	%	n	%	n	%	n	%	n	%	n	%	n	%
Male	2	1.5	46	33.6	10	7.3	0	0.0	58	42.3	2	25.0	60	41.4
Female	0	0.0	56	40.9	23	16.8	0	0.0	79	57.7	6	75.0	85	58.6
Total	2	1.5	102	74.5	33	24.1	0	0.0	137	100.0	8	100.0	145	100.0
MEAN LENGTH (mm)														
Male	510		563		579		na		564		576		564	
SE ^d	7.78		3.46		8.35		na		3.32		33.94		2.94	
Range	504	515	513	625	536	627	na	na	504	627	552	600	504	627
Female	na		559		568		na		562		558		562	
SE ^d	na		2.76		3.66		na		1.94		1.04		0.85	
Range	na	na	522	617	537	608	na	na	522	617	523	590	522	617
Combined	510		561		571		na		563		563		563	
SE ^d	7.78		2.16		3.56		na		1.93		9.86		1.88	
Range	504	515	513	625	536	627	na	na	504	627	523	600	504	627

^a Includes aged, sexed, and measured salmon.

^b Includes all unaged and measured salmon by sex.

^c For sexed samples, includes all salmon sampled whether aged and unaged.

^d Includes 3 Fall Chum salmon captured on August 9 (Drift # 92) that were excluded from the CPUE calculation.

^e SE calculated as $SD/\sqrt{n-1}$. Estimate may be slightly different than SE reported by ADF&G.

Table 5. Fall Chum Salmon sex ratio, age-class composition, by quartile and season total, Mountain Village Test Fish project, July 18–September 12, 2022.

Fall Chum Salmon Sampled										
Sex	Quartile									
	1		2		3		4		Total	
	18-Jul	10-Aug	11-Aug	17-Aug	18-Aug	23-Aug	24-Aug	12-Sep	18-Jul	12-Sep
	n ^a	%	n	%	n	%	n	%	n	%
Male	8	32.0	22	53.7	10	34.5	20	40.0	60	41.4
Female	17	68.0	19	46.3	19	65.5	30	60.0	85	58.6
unknown	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0
Total	25	100.0	41	100.0	29	100.0	50	100.0	145	100.0
Fall Chum Salmon Age										
Age	Quartile									
	1		2		3		4		Total	
	18-Jul	10-Aug	11-Aug	17-Aug	18-Aug	23-Aug	24-Aug	12-Sep	18-Jul	12-Sep
	n ^a	%	n	%	n	%	n	%	n	%
0.2	0	0.0	0	0.0	1	3.6	1	2.1	2	1.5
0.3	17	70.8	23	62.2	18	64.3	44	91.7	102	74.5
0.4	7	29.2	14	37.8	9	32.1	3	6.3	33	24.1
0.5	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0
Total Aged	24	96.0	37	90.2	28	96.6	48	96.0	137	94.5
unaged ^a	1	4.0	4	9.8	1	3.4	2	4.0	8	5.5
Grand Total	25	100.0	41	100.0	29	100.0	50	100.0	145	100.0
Age Error for Unaged Samples										
Age Error	Quartile									
	1		2		3		4		Total	
	18-Jul	10-Aug	11-Aug	17-Aug	18-Aug	23-Aug	24-Aug	12-Sep	18-Jul	12-Sep
	n	%	n	%	n	%	n	%	n	%
Missing	1	100.0	3	75.0	0	0.0	1	50.0	5	62.5
Regenerated	0	0.0	1	25.0	1	100.0	0	0.0	2	25.0
Inverted	0	0.0	0	0.0	0	0.0	1	50.0	1	12.5
Illegible	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0
Absorbed	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0
Total	1	100.0	4	100.0	1	100.0	2	100.0	8	100.0

^a Includes 3 Fall Chum Salmon captured on August 9 (Drift #92) that were not included in the CPUE calculations.

Table 6. Coho Salmon age, sex, and size summary, by brood year, for salmon caught and sampled in association with the Mountain Village Test Fish project, July 18 – September 12, 2022.

Brood Year (Age)												
	2018 (1.1)		2017 (2.1)		2016 (3.1)		Aged Sample ^{a,d}		Unaged sample ^b		Total sampled ^c	
NUMBER SAMPLED												
	#	%	#	%	#	%	#	%	#	%	#	%
Male	9	12.5	34	47.2	1	1.4	44	61.1	3	37.5	47	58.8
Female	6	8.3	21	29.2	1	1.4	28	38.9	5	62.5	33	41.3
Total	15	20.8	55	76.4	2	2.8	72	100.0	8	100.0	80	100.0
MEAN LENGTH (mm)												
Male	532		542		526		540		522		539	
See	7.98		3.93		na		3.44		15.95		3.33	
Range	482	568	464	583	526	526	464	583	468	553	464	583
Female	546		532		543		536		541		537	
SE ^e	10.09		5.04		na		4.34		11.28		3.99	
Range	482	568	464	583	526	526	464	583	468	553	464	583
Combined	537		539		535		538		534		538	
SE ^e	4.86		4.86		4.86		4.86		4.86		4.86	
Range	482	568	464	583	526	526	464	583	468	553	464	583

^a Includes aged, sexed, and measured salmon.

^b Includes all unaged and measured salmon by sex.

^c For sexed samples, includes all salmon sampled whether aged and unaged.

^d Includes 1 Coho Salmon captured on August 9 (Drift # 92) that was excluded from the CPUE calculation.

^e SE calculated as SD/sqrt(n-1). Estimate may be slightly different than SE reported by

ADF&G												
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Table 7. Coho Salmon sex ratio and age-class composition, by quartile and season total, Mountain Village Test Fish project, July 18–September 12, 2022.

Coho Salmon Sampled										
	Quartile									
	1		2		3		4		Total	
	18-Jul	20-Aug	21-Aug	25-Aug	26-Aug	30-Aug	31-Aug	12-Sep	18-Jul	12-Sep
Sex	n ^a	%	n	%	n	%	n	%	n	%
Male	15	65.2	8	57.1	10	47.6	14	63.6	47	58.8
Female	8	34.8	6	42.9	11	52.4	8	36.4	33	41.3
Total	23	100.0	14	100.0	21	100.0	22	100.0	80	100.0
Coho Salmon Age										
	Quartile									
	1		2		3		4		Total	
	18-Jul	20-Aug	21-Aug	25-Aug	26-Aug	30-Aug	31-Aug	12-Sep	18-Jul	12-Sep
Age	n ^a	%	n	%	n	%	n	%	n	%
1.1	3	14.3	2	14.3	2	11.8	8	40.0	15	20.8
2.1	18	85.7	12	85.7	13	76.5	12	60.0	55	76.4
3.1	0	0.0	0	0.0	2	11.8	0	0.0	2	2.8
Total Aged	21	100.0	14	100.0	17	100.0	20	100.0	72	100.0
unaged ^a	2	8.0	0	0.0	4	13.8	2	4.0	8	10.0
Grand Total	23	92.0	14	34.1	21	72.4	22	44.0	80	100.0
Age Error for Unaged Samples										
	Quartile									
	1		2		3		4		Total	
	18-Jul	20-Aug	21-Aug	25-Aug	26-Aug	30-Aug	31-Aug	12-Sep	18-Jul	12-Sep
Age Error	n	%	n	%	n	%	n	%	n	%
Missing	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0
Regenerated	0	0.0	4	50.0	3	37.5	1	12.5	8	100.0
Inverted	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0
Illegible	0	0.0	0	0.0	0	0.0	0	0.0	0	0.0
Total	0	0.0	4	50.0	3	37.5	1	12.5	8	100.0

^a Includes 1 Coho Salmon captured on August 9 (Drift #92) that were not included in the CPUE calculations.

FIGURES

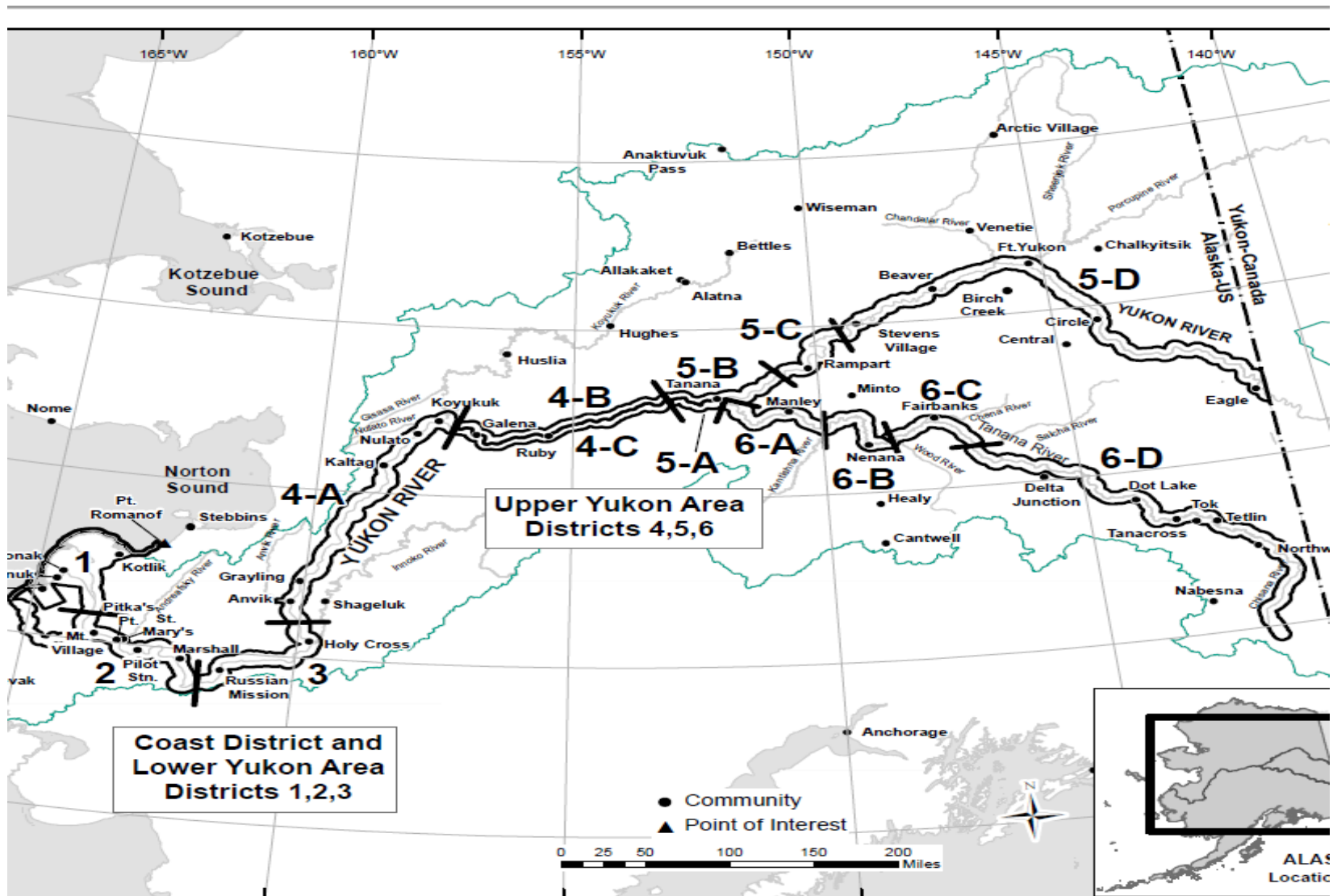


Figure 1. Map of the Yukon River drainage in Alaska showing management units.

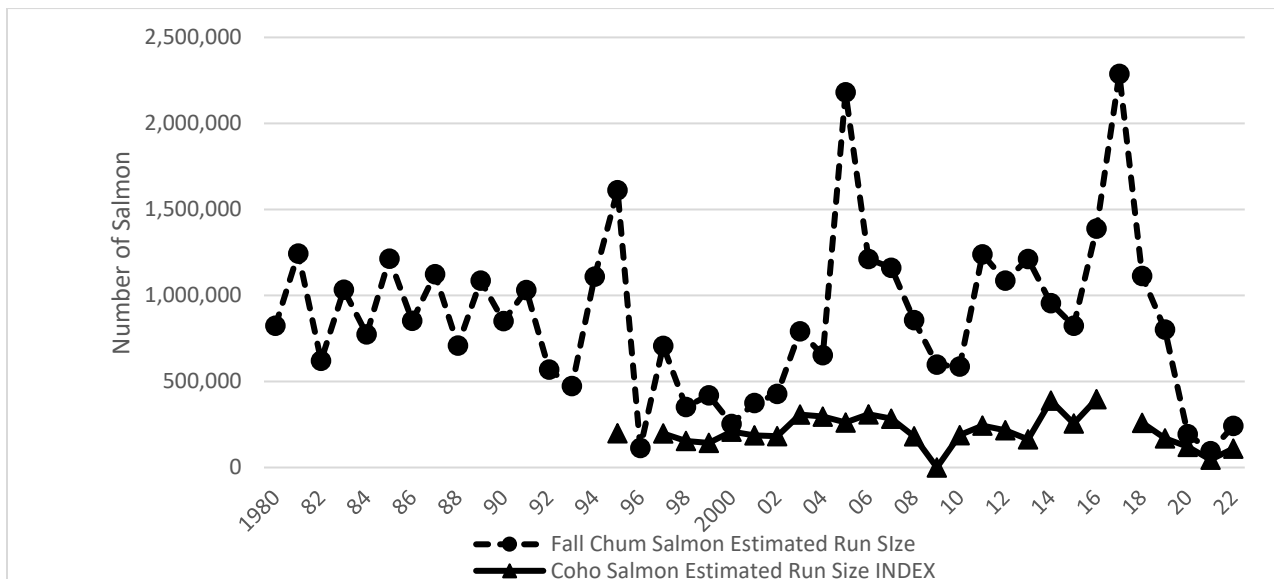


Figure 2. Estimated total run of Yukon River Fall Chum, 1980-2022, and the total run index of Yukon River Coho Salmon, 1980 – 1995 and 1997-2022.

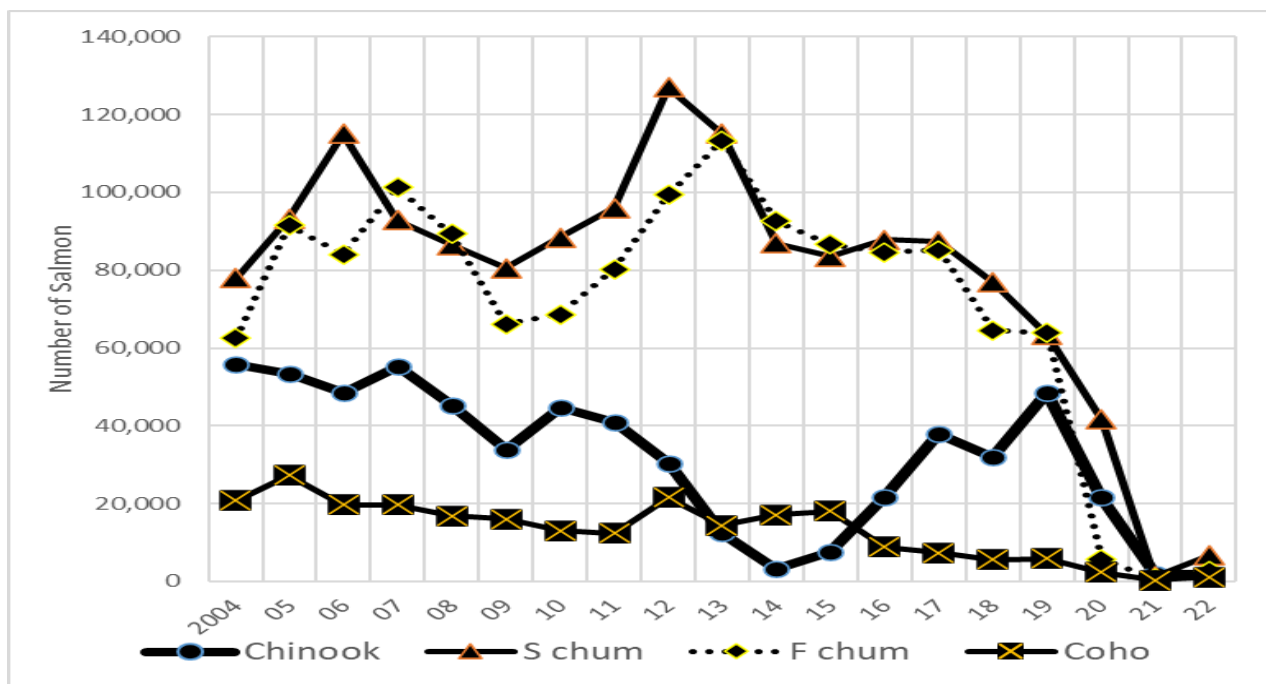


Figure 3. Subsistence salmon harvest, Yukon Area, Arctic-Yukon-Kuskokwim Region, 2004-2022.



Figure 4. Map of Yukon River in the vicinity of Mountain Village and the approximate location of the drift test fish Site 1, along the north bank of the Yukon River near Liberty Landing, Site 3 on the south side of the prominent sandbar and the test fishing sites, Sites 5 and 6, established during the 2019 season, located downriver of the community of Mountain Village.

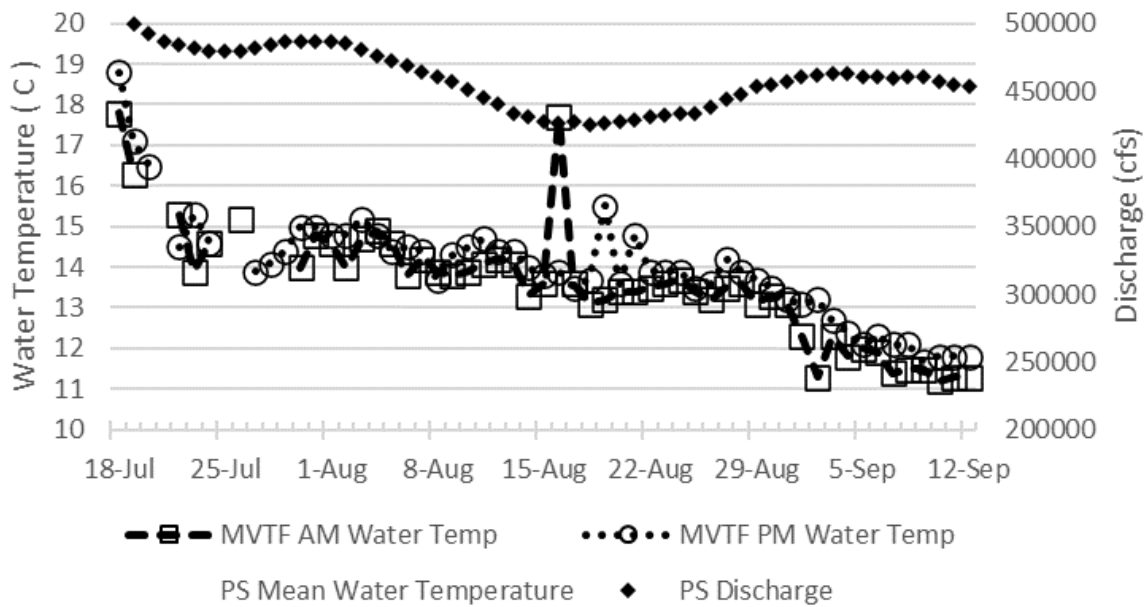


Figure 5. Instantaneous morning (AM) and evening (PM) water temperatures taken daily in association with the Mountain Village Test Fish (MVTF) project and mean water temperature and associated discharge (cfs) taken at the Pilot Station (PS) USGS gauging station (http://waterdata.usgs.gov/usa/nwis/uv?site_no=15565447), July 18 – September 12, 2022.

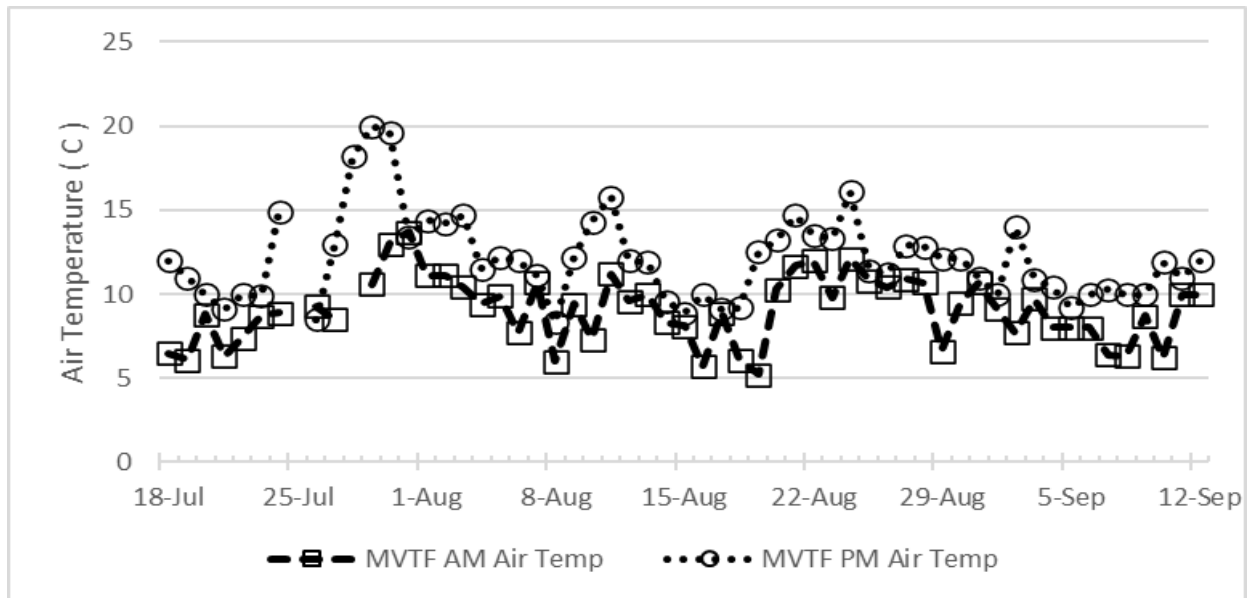


Figure 6. Instantaneous morning (AM) and evening (PM) air temperatures taken daily in association with the Mountain Village Test Fish (MVTF) project, July 18 – September 12, 2022.

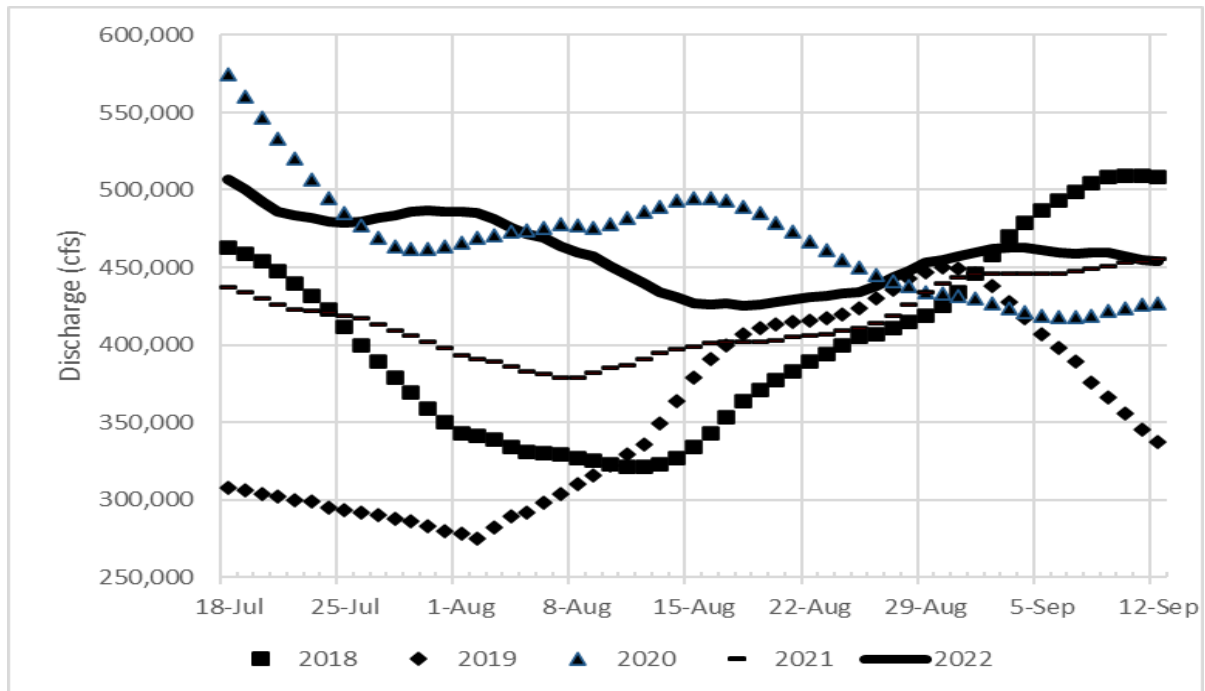


Figure 7. Yukon River discharge data collected at the Pilot Station gauging station at noon each day (http://waterdata.usgs.gov/usa/nwis/uv?site_no=15565447), July 18–September 12, 2018-2022.

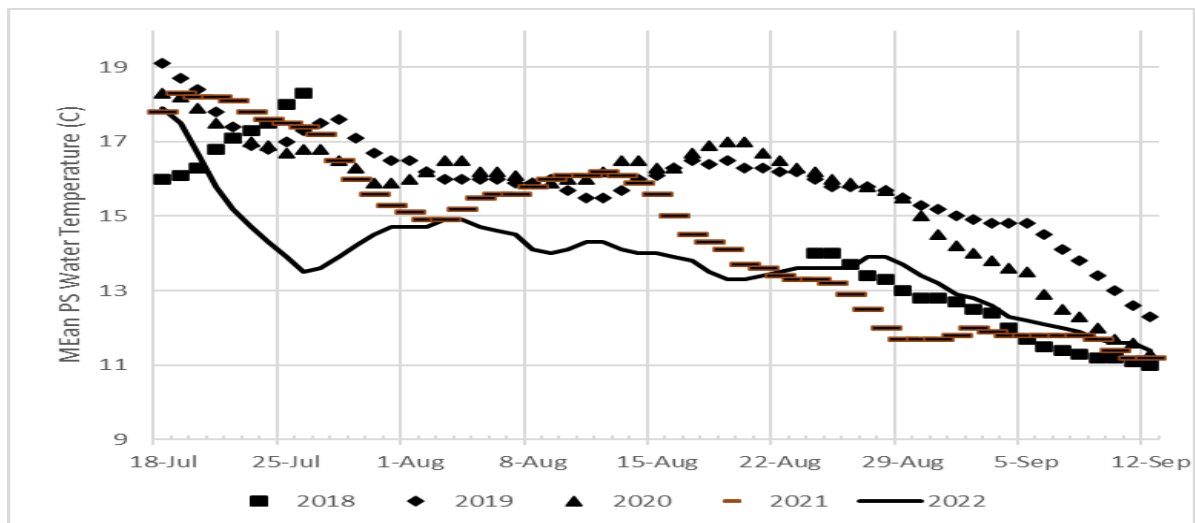


Figure 8. Yukon River daily mean water temperature collected at the Pilot Station gauging station (http://waterdata.usgs.gov/usa/nwis/uv?site_no=15565447), July 18–September 12, 2018-2022.

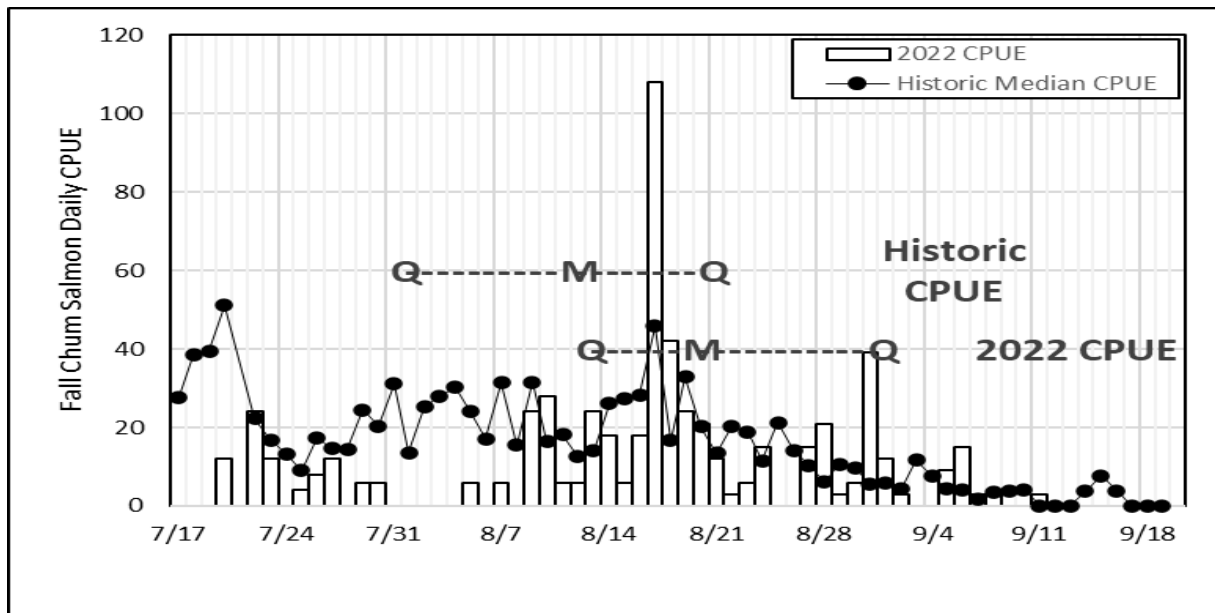


Figure 9. Historic, 1997–2012 and 2014–2021, median daily Fall Chum Salmon CPUE and the 2022 daily Fall Chum Salmon CPUE (with quartile “Q” and median “M” dates noted). Mountain Village Test Fish project.

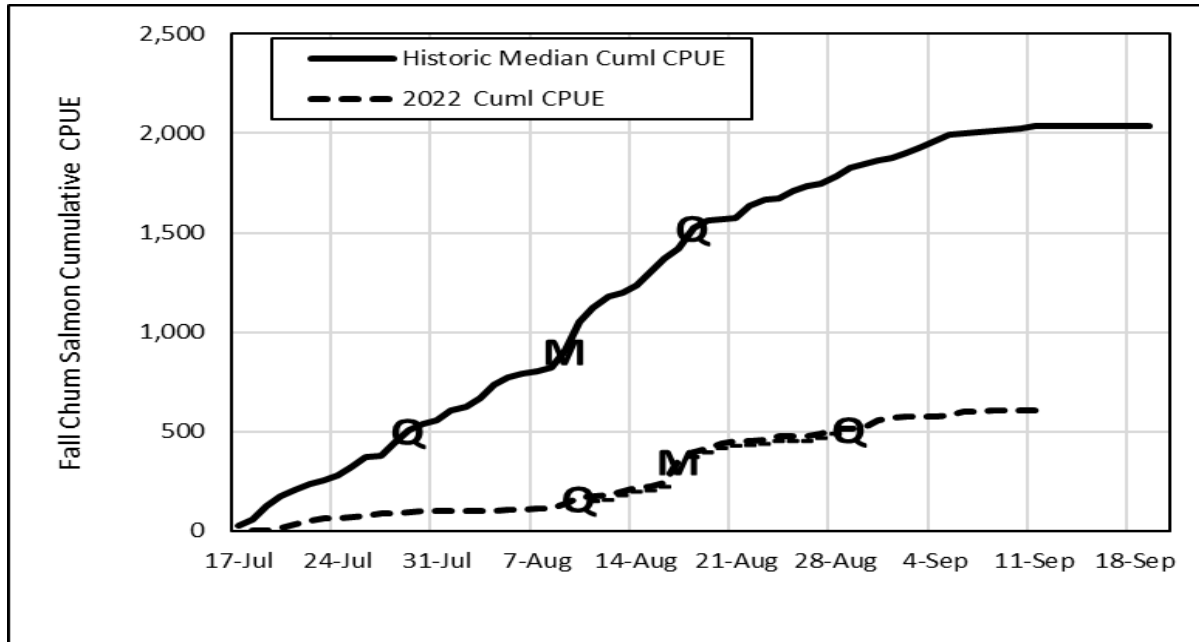


Figure 10. Historic (1997–2012 & 2014–2021) cumulative Fall Chum Salmon CPUE and the 2022 cumulative Fall Chum Salmon CPUE (with quartile “Q” and median “M” dates noted). Mountain Village Test Fish Project.

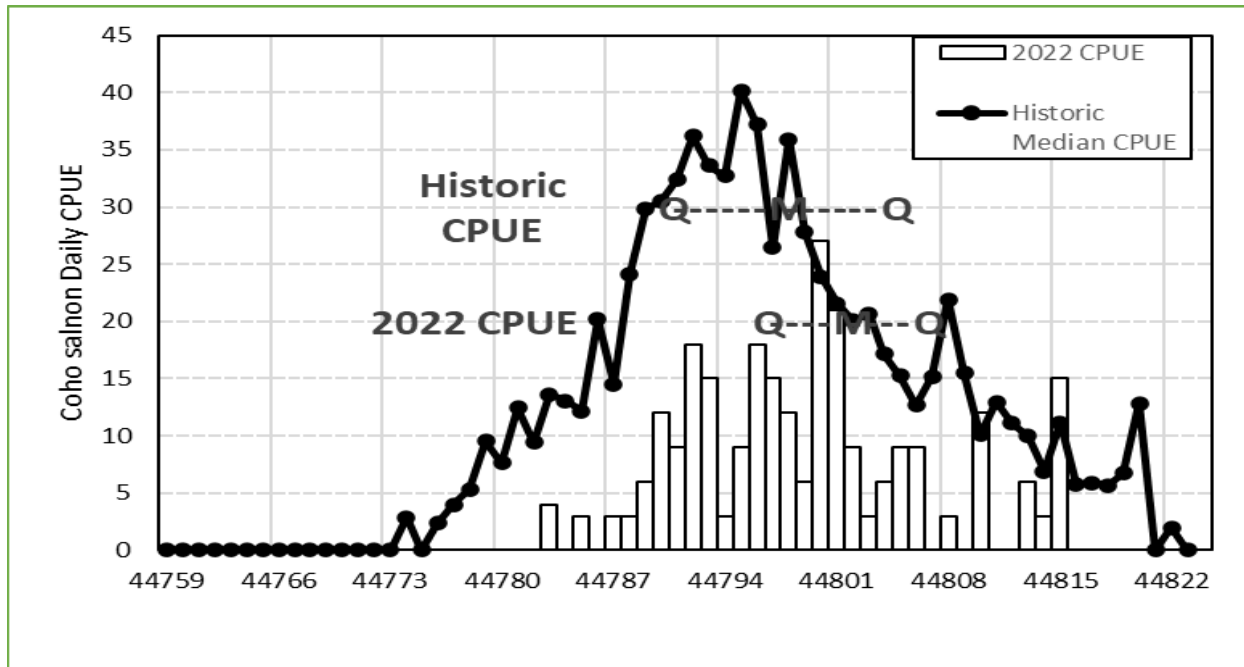


Figure 11. Historic, 1997–2012 and 2014–2021, median daily Coho Salmon CPUE and the 2022 daily Coho Salmon CPUE (with quartile “Q” and median “M” dates noted). Mountain Village Test Fish project.

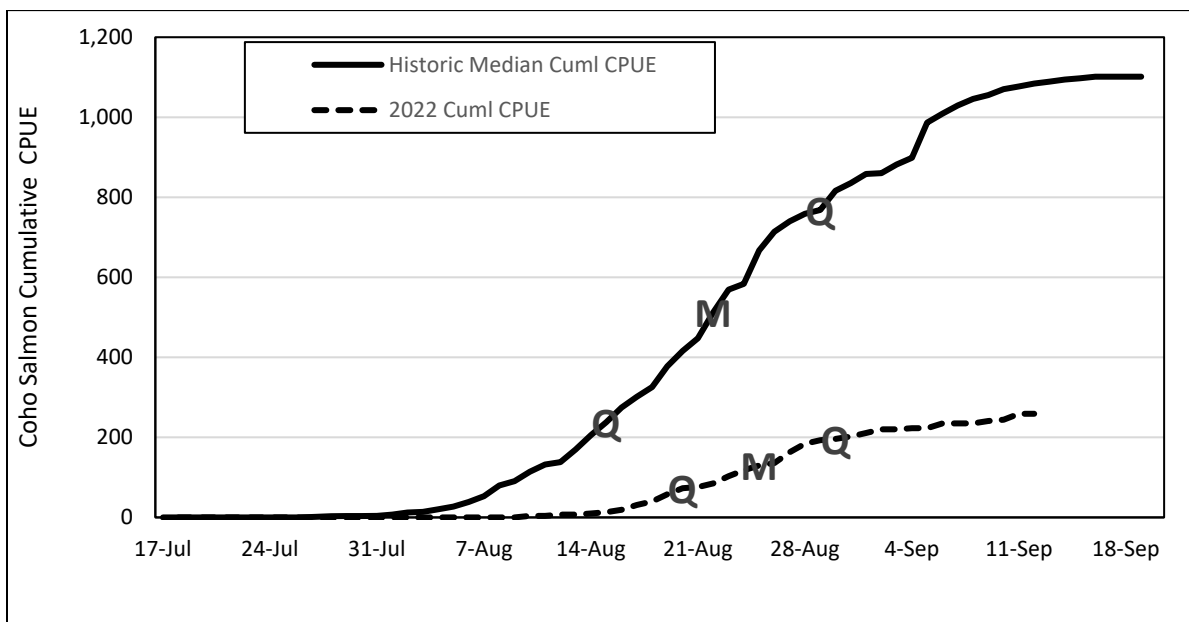


Figure 12. Historic (1997–2012 & 2014–2021) cumulative Coho Salmon CPUE and the 2022 cumulative Coho Salmon CPUE (with quartile “Q” and median “M” dates noted). Mountain Village Test Fish Project.

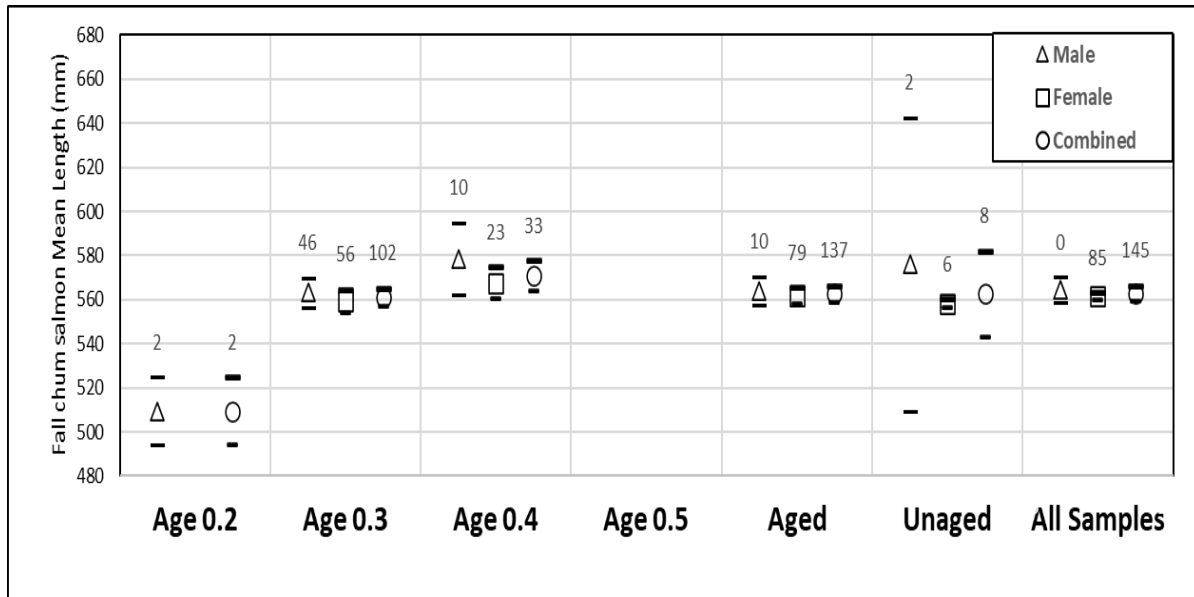


Figure 13. Mean length and corresponding 95% Confidence Interval for Fall Chum Salmon captured and sampled in association with the Mountain Village Test Fish project, July 18 – September 12, 2022. Numbers above the upper Confidence Interval bar indicate samples size associated for each mean length statistic.

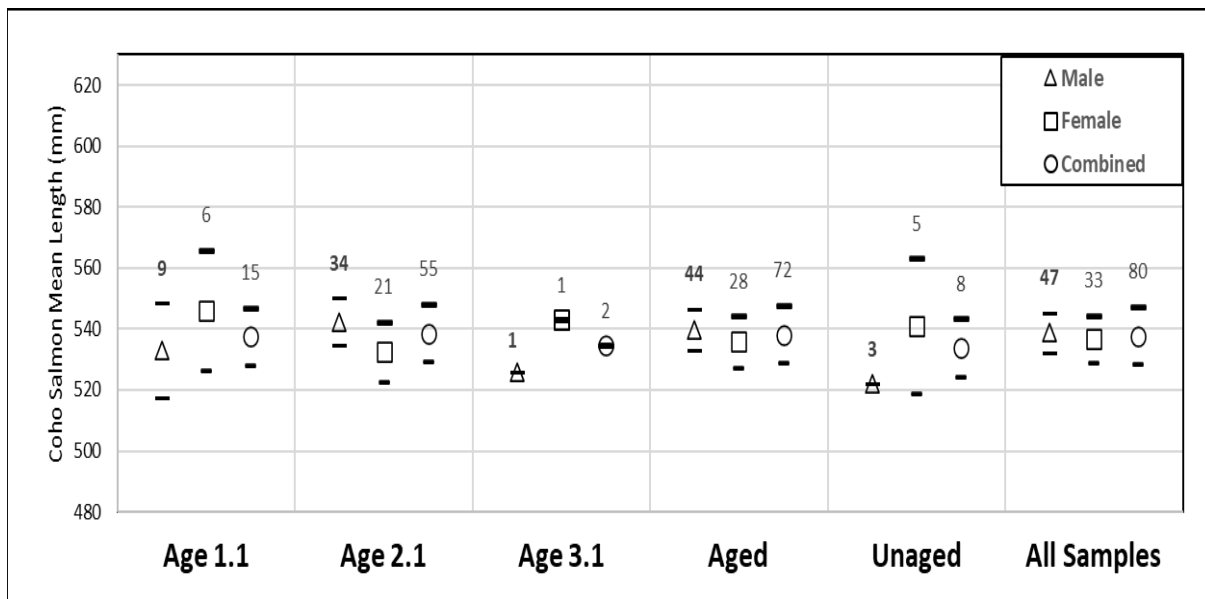


Figure 14. Mean length and corresponding 95% Confidence Interval for Coho Salmon captured and sampled in association with the Mountain Village Test Fish project, July 18 – September 12, 2022. Numbers above the upper Confidence Interval bar indicate samples size associated for each mean length statistic.

APPENDIX A:

Raw and mean fishing time, catch, and CPUE data by individual drifts, 2022.

Appendix A 1. Drift number, raw fishing time, mean fishing time, catch, and weighted Catch Per Unit Effort for Fall Chum and Coho Salmon, Mountain Village Test Fish project, July 18–September 12, 2022.

Drift No.	Date (2022)	Site No.	Fishing Time						Fall Chum Salmon				Coho Salmon				Other Fish Catch ^c
			Start Net Out	Net Full Out	Start Net In	Net Full Out	Mean Fishing Time	Drift Duration (Hr)	Retain	Release or Dropout	Catch ^a	Drift CPUE ^b	Retain	Release or Dropout	Catch ^a	Drift CPUE ^b	
1	18-Jul	5	8:00	8:01	8:19	8:20	19:00	0.317	0	0	0	0.00	0	0	0	0.00	
2	18-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
3	18-Jul	5	18:58	18:59	19:18	19:19	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
4	18-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
5	19-Jul	5	8:00	8:01	8:20	8:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
6	19-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
7	19-Jul	5	19:00	19:01	19:20	19:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
8	19-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
9	20-Jul	5	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
10	20-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
11	20-Jul	5	19:00	19:01	19:20	19:21	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
12	20-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
13	21-Jul	5	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
14	21-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
15	21-Jul	5	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
16	21-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
17	22-Jul	5	8:00	8:01	8:20	8:21	20:00	0.333	3	0	3	36.00	0	0	0	0.00	
18	22-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
19	22-Jul	5	19:00	19:01	19:20	19:21	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
20	22-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
21	23-Jul	5	8:00	8:01	8:20	8:21	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
22	23-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
23	23-Jul	5	19:00	19:01	19:20	19:21	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
24	23-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
25	24-Jul	5	8:00	8:01	8:20	8:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
26	24-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
27	24-Jul	5	19:00	19:01	19:20	19:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	1 PS
28	24-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
29	25-Jul	5	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
30	25-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
31	25-Jul	5	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
32	25-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
33	26-Jul	5	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
34	26-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
35	26-Jul	5	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
36	26-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
37	27-Jul	5	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
38	27-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
39	27-Jul	5	19:00	19:01	19:20	19:21	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
40	27-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	

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Drift No.	Date (2022)	Site No.	Fishing Time						Fall Chum Salmon				Coho Salmon				Other Fish Catch ^c
			Start Net Out	Net Full Out	Start Net In	Net Full Out	Mean Fishing Time	Drift Duration (Hr)	Retain	Release or Dropout	Catch ^a	Drift CPUE ^b	Retain	Release or Dropout	Catch ^a	Drift CPUE ^b	
41	28-Jul	5	8:00	8:01	8:20	8:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
42	28-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
43	28-Jul	5	19:00	19:01	19:20	19:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
44	28-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
45	29-Jul	5	8:00	8:01	8:20	8:21	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
46	29-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
47	29-Jul	5	19:00	19:01	19:20	19:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
48	29-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
49	30-Jul	5	8:00	8:01	8:20	8:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
50	30-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
51	30-Jul	5	19:00	19:01	19:20	19:21	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
52	30-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
53	31-Jul	5	8:00	8:01	8:20	8:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
54	31-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
55	31-Jul	5	19:00	19:01	19:20	19:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
56	31-Jul	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
57	1-Aug	5	8:00	8:01	8:20	8:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
58	1-Aug	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
59	1-Aug	5	19:00	19:01	19:20	19:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
60	1-Aug	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
61	2-Aug	5	8:00	8:01	8:20	8:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
62	2-Aug	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
63	2-Aug	5	19:00	19:01	19:20	19:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
64	2-Aug	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
65	3-Aug	5	8:00	8:01	8:20	8:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
66	3-Aug	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
67	3-Aug	5	19:00	19:01	19:20	19:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	1 PS
68	3-Aug	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
69	4-Aug	5	8:00	8:01	8:20	8:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
70	4-Aug	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
71	4-Aug	5	19:00	19:01	19:20	19:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
72	4-Aug	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
73	5-Aug	5	8:00	8:01	8:20	8:21	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
74	5-Aug	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
75	5-Aug	5	19:00	19:01	19:20	19:21	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
76	5-Aug	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
77	6-Aug	5	8:01	8:02	8:20	8:21	19:00	0.317	0	0	0	0.00	0	0	0	0.00	
78	6-Aug	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
79	6-Aug	5	18:54	18:55	19:14	19:15	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
80	6-Aug	6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	

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				Fishing Time						Fall Chum Salmon				Coho Salmon						
Drift No.	Date (2022)		Site No.	Start Net Out	Net Full Out	Start Net In	Net Full Out	Mean Fishing Time	Drift Duration (Hr)		Retain	Release or Dropout	Catch ^a	Drift CPUE ^b	Retain	Release or Dropout	Catch ^a	Drift CPUE ^b	Other Fish Catch ^c	
81	7-Aug		5	8:54	8:55	9:14	9:15	20:00	0.333		0	0	0	0.00	0	0	0	0.00		
82	7-Aug	d	6	na	na	na	na	0:00	0.000		0	0	0	0.00	0	0	0	0.00		
83	7-Aug		5	18:51	18:52	19:11	19:12	20:00	0.333		0	1	1	12.00	0	0	0	0.00		
84	7-Aug	d	6	na	na	na	na	0:00	0.000		0	0	0	0.00	0	0	0	0.00		
85	8-Aug		5	7:58	7:59	8:18	8:19	20:00	0.333		0	0	0	0.00	0	0	0	0.00		
86	8-Aug	d	6	na	na	na	na	0:00	0.000		0	0	0	0.00	0	0	0	0.00		
87	8-Aug	e	5	na	na	na	na	0:00	0.000		0	0	0	0.00	0	0	0	0.00		
88	8-Aug	e	6	na	na	na	na	0:00	0.000		0	0	0	0.00	0	0	0	0.00		
89	9-Aug		5	7:52	7:53	8:12	8:13	20:00	0.333		4	0	4	48.00	0	0	0	0.00		
90	9-Aug	d	6	na	na	na	na	0:00	0.000		0	0	0	0.00	0	0	0	0.00		
91	9-Aug		5	18:48	18:49	19:08	19:09	20:00	0.333		0	0	0	0.00	0	0	0	0.00		
92	9-Aug	d g	6	na	na	na	na	na	na		3	0	3	na	1	0	1	na		
93	10-Aug		5	9:36	9:37	9:56	9:57	20:00	0.333		3	0	3	36.00	1	0	1	12.00		
94	10-Aug	d	6	na	na	na	na	0:00	0.000		0	0	0	0.00	0	0	0	0.00		
95	10-Aug		5	18:48	18:49	19:08	19:09	20:00	0.333		4	0	4	48.00	0	0	0	0.00		
96	10-Aug		6	19:16	19:17	19:36	19:37	20:00	0.333		0	0	0	0.00	0	0	0	0.00		
97	11-Aug		5	11:13	11:14	11:33	11:34	20:00	0.333		2	0	2	24.00	0	0	0	0.00		
98	11-Aug		6	11:38	11:39	11:58	11:59	20:00	0.333		0	0	0	0.00	0	0	0	0.00		
99	11-Aug		5	18:45	18:46	19:05	19:06	20:00	0.333		0	0	0	0.00	0	0	0	0.00		
100	11-Aug		6	19:13	19:14	19:33	19:34	20:00	0.333		0	0	0	0.00	0	0	0	0.00		
101	12-Aug		5	9:39	9:40	9:59	10:00	20:00	0.333		0	0	0	0.00	0	0	0	0.00		
102	12-Aug		6	10:15	10:16	10:35	10:36	20:00	0.333		0	0	0	0.00	0	0	0	0.00		
103	12-Aug		5	19:12	19:13	19:32	19:33	20:00	0.333		1	0	1	12.00	1	0	1	12.00		
104	12-Aug		6	19:39	19:40	19:59	20:00	20:00	0.333		1	0	1	12.00	0	0	0	0.00		
105	13-Aug		5	9:35	9:36	9:55	9:56	20:00	0.333		2	0	2	24.00	0	0	0	0.00		
106	13-Aug		6	10:02	10:03	10:22	10:23	20:00	0.333		3	0	3	36.00	0	0	0	0.00		
107	13-Aug		5	18:51	18:52	19:11	19:12	20:00	0.333		1	0	1	12.00	0	0	0	0.00		
108	13-Aug		6	19:19	19:20	19:39	19:40	20:00	0.333		2	0	2	24.00	0	0	0	0.00		
109	14-Aug		5	7:58	7:59	8:18	8:19	20:00	0.333		1	0	1	12.00	0	0	0	0.00		
110	14-Aug		6	8:29	8:30	8:49	8:50	20:00	0.333		4	0	4	48.00	0	0	0	0.00		
111	14-Aug		5	18:44	18:45	19:04	19:05	20:00	0.333		0	0	0	0.00	1	0	1	12.00		
112	14-Aug		6	19:14	19:15	19:34	19:35	20:00	0.333		1	0	1	12.00	0	0	0	0.00		
113	15-Aug		5	7:57	7:58	8:17	8:18	20:00	0.333		1	0	1	12.00	0	0	0	0.00		
114	15-Aug		6	8:26	8:27	8:46	8:47	20:00	0.333		0	0	0	0.00	1	0	1	12.00		
115	15-Aug		5	18:44	18:45	19:04	19:05	20:00	0.333		0	0	0	0.00	0	0	0	0.00		
116	15-Aug		6	19:10	19:11	19:30	19:31	20:00	0.333		1	0	1	12.00	0	0	0	0.00		
117	16-Aug		5	8:04	8:05	8:24	8:25	20:00	0.333		0	0	0	0.00	0	0	0	0.00		
118	16-Aug		6	8:31	8:32	8:51	8:52	20:00	0.333		2	0	2	24.00	0	0	0	0.00		
119	16-Aug		5	18:44	18:45	19:04	19:05	20:00	0.333		0	0	0	0.00	0	0	0	0.00		
120	16-Aug		6	19:14	19:15	19:34	19:35	20:00	0.333		4	0	4	48.00	2	0	2	24.00		
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Drift No.	Date (2022)	Site No.	Fishing Time						Fall Chum Salmon				Coho Salmon				Other Fish Catch ^c
			Start Net Out	Net Full Out	Start Net In	Net Full Out	Mean Fishing Time	Drift Duration (Hr)	Retain	Release or Dropout	Catch ^a	Drift CPUE ^b	Retain	Release or Dropout	Catch ^a	Drift CPUE ^b	
121	17-Aug	5	8:58	8:59	9:18	9:19	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
122	17-Aug	6	9:27	9:28	9:47	9:48	20:00	0.333	15	3	18	216.00	2	0	2	24.00	
123	17-Aug	^e 5	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
124	17-Aug	^e 6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
125	18-Aug	5	8:04	8:05	8:24	8:25	20:00	0.333	1	0	1	12.00	1	0	1	12.00	
126	18-Aug	6	8:34	8:35	8:54	8:55	20:00	0.333	6	0	6	72.00	0	0	0	0.00	
127	18-Aug	5	18:49	18:50	19:09	19:10	20:00	0.333	2	1	3	36.00	0	0	0	0.00	
128	18-Aug	6	19:17	19:18	19:37	19:38	20:00	0.333	3	1	4	48.00	2	0	2	24.00	
129	19-Aug	5	7:58	7:59	8:18	8:19	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
130	19-Aug	6	8:25	8:26	8:45	8:46	20:00	0.333	3	1	4	48.00	0	0	0	0.00	
131	19-Aug	5	18:58	18:59	19:18	19:19	20:00	0.333	2	0	2	24.00	3	0	3	36.00	
132	19-Aug	6	19:29	19:30	19:49	19:50	20:00	0.333	3	0	3	36.00	3	0	3	36.00	
133	20-Aug	5	8:29	8:30	8:49	8:50	20:00	0.333	3	0	3	36.00	3	0	3	36.00	
134	20-Aug	6	8:58	8:59	9:18	9:19	20:00	0.333	2	0	2	24.00	0	0	0	0.00	
135	20-Aug	5	19:04	19:05	19:24	19:25	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
136	20-Aug	6	19:31	19:32	19:51	19:52	20:00	0.333	2	0	2	24.00	2	0	2	24.00	
137	21-Aug	5	8:09	8:10	8:29	8:30	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
138	21-Aug	6	8:35	8:36	8:55	8:56	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
139	21-Aug	5	18:02	18:03	18:22	18:23	20:00	0.333	2	0	2	24.00	0	0	0	0.00	
140	21-Aug	6	19:31	19:32	19:51	19:52	20:00	0.333	0	1	1	12.00	1	0	1	12.00	
141	22-Aug	5	7:57	7:58	8:17	8:18	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
142	22-Aug	6	8:23	8:24	8:43	8:44	20:00	0.333	0	1	1	12.00	0	1	1	12.00	1 SF
143	22-Aug	5	18:56	18:57	19:16	19:17	20:00	0.333	0	0	0	0.00	1	0	1	12.00	
144	22-Aug	6	19:23	19:24	19:43	19:44	20:00	0.333	0	0	0	0.00	1	0	1	12.00	
145	23-Aug	5	10:13	10:14	10:33	10:34	20:00	0.333	0	0	0	0.00	1	1	2	24.00	
146	23-Aug	6	10:40	10:41	11:00	11:01	20:00	0.333	0	0	0	0.00	3	1	4	48.00	
147	23-Aug	5	18:56	18:57	19:16	19:17	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
148	23-Aug	6	19:22	19:23	19:42	19:43	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
149	24-Aug	5	7:58	7:59	8:18	8:19	20:00	0.333	0	1	1	12.00	1	0	1	12.00	
150	24-Aug	6	8:26	8:27	8:46	8:47	20:00	0.333	2	0	2	24.00	1	0	1	12.00	
151	24-Aug	5	18:47	18:48	19:07	19:08	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
152	24-Aug	6	19:16	19:17	19:36	19:37	20:00	0.333	2	0	2	24.00	3	0	3	36.00	
153	25-Aug	^e 5	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
154	25-Aug	^e 6	na	na	na	na	0:00	0.000	0	0	0	0.00	0	0	0	0.00	
155	25-Aug	5	18:40	18:41	19:00	19:01	20:00	0.333	0	0	0	0.00	1	0	1	12.00	
156	25-Aug	6	19:06	19:07	19:26	19:27	20:00	0.333	0	0	0	0.00	1	0	1	12.00	
157	26-Aug	5	7:43	7:44	8:03	8:04	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
158	26-Aug	6	8:08	8:09	8:28	8:29	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
159	26-Aug	5	18:43	18:44	19:03	19:04	20:00	0.333	0	0	0	0.00	1	0	1	12.00	
160	26-Aug	5	19:10	19:11	19:30	19:31	20:00	0.333	0	0	0	0.00	1	0	1	12.00	1 RS

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Drift No.	Date (2022)	Site No.	Fishing Time						Fall Chum Salmon				Coho Salmon				Other Fish Catch ^c
			Start Net Out	Net Full Out	Start Net In	Net Full Out	Mean Fishing Time	Drift Duration (Hr)	Retain	Release or Dropout	Catch ^a	Drift CPUE ^b	Retain	Release or Dropout	Catch ^a	Drift CPUE ^b	
161	27-Aug	5	7:50	7:51	8:10	8:11	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
162	27-Aug	6	8:17	8:18	8:37	8:38	20:00	0.333	4	0	4	48.00	1	0	1	12.00	
163	27-Aug	5	18:44	18:45	19:04	19:05	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
164	27-Aug	6	19:10	19:11	19:30	19:31	20:00	0.333	1	0	1	12.00	8	0	8	96.00	
165	28-Aug	5	7:48	7:49	8:08	8:09	20:00	0.333	1	0	1	12.00	2	0	2	24.00	
166	28-Aug	6	8:15	8:16	8:35	8:36	20:00	0.333	4	0	4	48.00	1	0	1	12.00	
167	28-Aug	5	18:51	18:52	19:11	19:12	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
168	28-Aug	6	19:18	19:19	19:38	19:39	20:00	0.333	1	0	1	12.00	4	0	4	48.00	
169	29-Aug	5	7:49	7:50	8:09	8:10	20:00	0.333	0	0	0	0.00	2	0	2	24.00	
170	29-Aug	6	8:16	8:17	8:36	8:37	20:00	0.333	0	0	0	0.00	1	0	1	12.00	
171	29-Aug	5	18:47	18:48	19:07	19:08	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
172	29-Aug	6	19:13	19:14	19:33	19:34	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
173	30-Aug	5	7:49	7:50	8:09	8:10	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
174	30-Aug	6	8:15	8:16	8:35	8:36	20:00	0.333	1	0	1	12.00	1	0	1	12.00	
175	30-Aug	5	18:50	18:51	19:10	19:11	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
176	30-Aug	6	19:17	19:18	19:37	19:38	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
177	31-Aug	5	7:52	7:53	8:12	8:13	20:00	0.333	0	0	0	0.00	1	0	1	12.00	
178	31-Aug	6	8:19	8:20	8:39	8:40	20:00	0.333	11	0	11	132.00	1	0	1	12.00	
179	31-Aug	5	18:44	18:45	19:04	19:05	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
180	31-Aug	6	19:11	19:12	19:31	19:32	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
181	1-Sep	5	7:59	8:00	8:19	8:20	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
182	1-Sep	6	8:25	8:26	8:45	8:46	20:00	0.333	4	0	4	48.00	1	0	1	12.00	
183	1-Sep	5	18:30	18:31	18:50	18:51	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
184	1-Sep	6	18:56	18:57	19:16	19:17	20:00	0.333	0	0	0	0.00	2	0	2	24.00	
185	2-Sep	5	7:52	7:53	8:12	8:13	20:00	0.333	0	0	0	0.00	1	0	1	12.00	
186	2-Sep	6	8:18	8:19	8:38	8:39	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
187	2-Sep	5	18:12	18:13	18:32	18:33	20:00	0.333	0	0	0	0.00	1	0	1	12.00	
188	2-Sep	6	18:39	18:40	18:59	19:00	20:00	0.333	0	0	0	0.00	1	0	1	12.00	
189	3-Sep	5	7:48	7:49	8:08	8:09	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
190	3-Sep	6	8:15	8:16	8:35	8:36	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
191	3-Sep	5	18:56	18:57	19:16	19:17	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
192	3-Sep	6	19:26	19:27	19:46	19:47	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
193	4-Sep	5	7:53	7:54	8:13	8:14	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
194	4-Sep	6	8:19	8:20	8:39	8:40	20:00	0.333	0	0	0	0.00	1	0	1	12.00	
195	4-Sep	5	18:16	18:17	18:36	18:37	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
196	4-Sep	6	18:42	18:43	19:02	19:03	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
197	5-Sep	5	7:45	7:46	8:05	8:06	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
198	5-Sep	6	8:12	8:13	8:32	8:33	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
199	5-Sep	5	18:11	18:12	18:31	18:32	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
200	5-Sep	6	18:37	18:38	18:57	18:58	20:00	0.333	2	0	2	24.00	0	0	0	0.00	

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Drift No.	Date (2022)	Site No.	Fishing Time						Fall Chum Salmon				Coho Salmon				Other Fish Catch ^c
			Start Net Out	Net Full Out	Start Net In	Net Full Out	Mean Fishing Time	Drift Duration (Hr)	Retain	Release or Dropout	Catch ^a	Drift CPUE ^b	Retain	Release or Dropout	Catch ^a	Drift CPUE ^b	
201	6-Sep	5	8:13	8:14	8:33	8:34	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
202	6-Sep	6	8:40	8:41	9:00	9:01	20:00	0.333	1	0	1	12.00	1	0	1	12.00	
203	6-Sep	5	18:14	18:15	18:34	18:35	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
204	6-Sep	6	18:41	18:42	19:01	19:02	20:00	0.333	3	0	3	36.00	3	0	3	36.00	
205	7-Sep	5	8:23	8:24	8:43	8:44	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
206	7-Sep	6	8:50	8:51	9:10	9:11	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
207	7-Sep	5	18:11	18:12	18:31	18:32	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
208	7-Sep	6	18:38	18:39	18:58	18:59	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
209	8-Sep	5	8:23	8:24	8:43	8:44	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
210	8-Sep	6	8:50	8:51	9:10	9:11	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
211	8-Sep	5	18:12	18:13	18:32	18:33	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
212	8-Sep	6	18:39	18:40	18:59	19:00	20:00	0.333	1	0	1	12.00	0	0	0	0.00	
213	9-Sep	5	8:25	8:26	8:45	8:46	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
214	9-Sep	6	8:52	8:53	9:12	9:13	20:00	0.333	0	0	0	0.00	2	0	2	24.00	
215	9-Sep	5	18:19	18:20	18:39	18:40	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
216	9-Sep	6	18:46	18:47	19:06	19:07	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
217	10-Sep	5	8:23	8:24	8:43	8:44	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
218	10-Sep	6	8:50	8:51	9:10	9:11	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
219	10-Sep	5	18:10	18:11	18:30	18:31	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
220	10-Sep	6	18:37	18:38	18:57	18:58	20:00	0.333	0	0	0	0.00	1	0	1	12.00	
221	11-Sep	5	8:18	8:19	8:38	8:39	20:00	0.333	0	0	0	0.00	1	0	1	12.00	
222	11-Sep	6	8:44	8:45	9:04	9:05	20:00	0.333	0	0	0	0.00	1	0	1	12.00	
223	11-Sep	5	18:20	18:21	18:40	18:41	20:00	0.333	0	0	0	0.00	2	0	2	24.00	
224	11-Sep	6	18:47	18:48	19:07	19:08	20:00	0.333	1	0	1	12.00	1	0	1	12.00	
225	12-Sep	5	8:19	8:20	8:39	8:40	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
226	12-Sep	6	8:46	8:47	9:06	9:07	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
227	12-Sep	5	18:14	18:15	18:34	18:35	20:00	0.333	0	0	0	0.00	0	0	0	0.00	
228	12-Sep	6	18:41	18:42	19:01	19:02	20:00	0.333	0	0	0	0.00	0	0	0	0.00	

^a Catch includes those fish retained, released, or that have dropped out of the net.

^b Weighted CPUE expressed as catch per 100 net fathoms per hour. ADF&G values.

^c Other fish include: KS=Chinook Salmon; SO=Sockeye Salmon; SF=sheefish; WF=whitefish

^d Site 6 was not fished because of extremely high water covering the sand bar.

^e Test fishing drifts were cancelled because of hazardous river water conditions.

^f Test fishing did not occur because of boat mechanical issue.

^g Net caught on a snag and remained overnight. Catch included 3 Fall Chum Salmon and 1 Coho salmon that were not included in CPUE calculations but were sampled.